Deconstructing Wrong-Way Risk in the Presence of Initial Margin

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Candidate Number: PWHF0

A dissertation submitted for the degree of

Masters of Science

In

Financial Mathematics

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September, 2025

Disclaimer

I, Dominic Tang, confirm that the work presented in this thesis is my own. Where information has been derived from other sources, I confirm that this has been indicated in the thesis. Generative AI was used to help with proofreading parts of the dissertation.

Abstract

The post-2008 regulatory framework, centered on Variation and Initial Margin, was designed to mitigate counterparty default risk in derivatives markets. However, the collapse of Archegos Capital Management in 2021 exposed a fundamental weakness in this framework: leveraged Wrong-Way Risk (WWR), where collateral via initial margin proves insufficient precisely when exposure is highest. While modern frameworks, such as that proposed by Pykhtin, have been developed to quantify this risk, we are investigating on whether the classic WWR model includes the "double counting" problem: layering an explicit WWR model on top of market credit spreads that already contain an implicit risk premium.

This dissertation tests the hypothesis that market premium serves as a partial buffer mechanism against counterparty default risk and aims to quantify the true residual credit valuation adjustment (CVA) under different levels of Initial margin (IM). This value is then used to investigate whether initial margin is able to cover potential losses as intended. Through experiments, we decompose market credit pricing: first, we calibrate the "risk-free" hazard rate from the CVA of an unsecured portfolio, then apply this rate to secured portfolios based on three stochastic processes – geometric Brownian motion, the Heston model, and the Merton model. The results demonstrate that, despite the market credit premium explaining the bulk of the WWR effect, a significant residual risk persists. The research reveals that, when accounting for the WWR effect, the CVA for collateralised trades is still several times higher than when ignoring this effect. This paper ultimately proposes a more refined CVA calculation framework that avoids both the underestimation of conventional models and the overestimation of simple WWR models, providing a practical method for pricing the true residual risk to counterparties in leveraged trades.

This dissertation was completed under the supervision of Mr. Chia Chiang Tan.

Contents

1 Introduction

The 2008 global financial crisis was a stark reminder of the systemic risk triggered by counterparty default – how the failure of a single institution could set off a domino effect across interconnected financial systems. In response, global regulators built a new framework centred on Variation Margin and Initial Margin, designed to strengthen derivatives markets to withstand such collapses. For years, this collateral framework appeared robust, fostering a sense of security that the core counterparty risk mechanism was under control. However, the collapse of Archegos Capital Management in March 2021 shattered this perception. This exposed significant vulnerabilities in the original regulatory framework, and resulting in counterparty losses which far exceeded what traditional risk models had predicted.

The mandatory initial margin (IM) posting for non-centrally cleared derivatives are driven by regulations such as the BSBC-IOSCO framework [3] and implemented through models like the Standardised Initial Margin Model (SIMM). This represents a significant shift in the risk management world. Within this new margin environment, initial margin is designed as a powerful mitigating mechanism, substantially reducing expected exposure and consequently diminishing credit valuation adjustment (CVA) for the vast majority of trades. This has fostered a widespread perception that counterparty risk within portfolios with IM has been fundamentally resolved. The core research of this paper is based on this new world, assuming that the CVA in uncollateralised portfolios is efficiently priced by the market and explores the nature and scale of residual risk in an environment where margin systems, including initial margin and variation margin, are widely implemented.

The Archegos default showcases a troublesome form of counterparty risk known as leveraged Wrong-Way Risk (WWR). This phenomenon occurs when the probability of a counterparty default exhibits a significant positive correlation with the size of credit exposure to that counterparty, creating a perfect storm where the protection is least effective when it is most needed. The mechanism of Archegos' collapse can be attributed to two correlated outcomes triggered by a single, severe market shock, leading to a sharp decline in the value of the firm's highly concentrated investment portfolio: [2]

1. The market shock caused a rapid increase in the credit exposure of counterparties to Archegos.

2. Simultaneously, the shock directly triggered Archegos' default, as the firm was unable to meet the resulting massive margin calls.

This phenomenon demonstrates that for highly leveraged entities, market risk and credit risk are not independent variables – a flawed assumption in many traditional models. Therefore, this paper adopts the framework proposed by Pykhtin to quantify the extent of this potential underestimation of credit valuation adjustment (CVA). While the model indicates CVA is significantly underestimated, this study critically examines that conclusion. The core objective is to explore whether this substantial underestimation is as severe as the model suggests, or whether the market has already priced in this tail risk through other mechanisms. Such mechanisms include the overestimation of market-implied default probabilities over historical default rates.

Specifically, this dissertation tests the hypothesis that the known premium between the default probability implicitly signalled by the market (derived from tools such as Credit Default Swaps) and the historically observed default probability may already capture risks such as WWR. This question is particularly crucial in the context of trades protected by Initial Margin (IM). While IM is designed to cover the vast majority of potential future losses through collateral, a residual tail risk remains. It is in this tail-risk scenario where the leveraged WWR is the most dangerous. By comparing the credit value adjustment (CVA) values adjusted for WWR under different IM levels with those based on the market-implied default hazard rate, this study aims to quantify the true CVA gap. In particular, we will focus on the effects of WWR on the CVA for IM at 95% and 99% respectively, and examine whether the IM is as effective at covering potential losses as intended. This analysis aims to clarify whether models incorporating leveraged WWR identify genuinely underpriced risks or merely make explicit risk premiums already implied and priced by the market.

2 Model Specification

2.1 Portfolio Instruments and Pricing Functions

The portfolio subject to counterparty risk assumes only a single European-style derivative contract based on the underlying asset S_t. The value of this derivative at any time t prior to its maturity date T is denoted by $V(S_t, t)$. We analyse its robustness across different risk exposure profiles using the following three standard contract types: Call, Put and Forward.

This analysis focuses on single-name derivatives – specifically European call options, put options, and forward contracts. This selection was made based on the theoretical objectives and empirical context of this paper. The Archegos case involved trend positions that are highly concentrated in a small number of individual equities. Therefore, modelling using vanilla equity options and forward contracts is the most direct and relevant approach to simulating Archegos-type risk exposures.

A classic real-world example of Wrong-Way Risk (WWR) involves foreign exchange (FX) exposure in contracts with counterparties in developing economies. Consider a contract where a counterparty, whose revenues and expenses are primarily in their local currency, is obligated to pay a fixed amount in a stronger currency like USD. If that developing economy suffers a severe crisis, two correlated events are likely to unfold. First, the crisis would weaken the local currency against the USD, dramatically increasing the counterparty's debt burden in their local currency. Second, the same crisis would simultaneously damage their underlying business, crippling their ability to generate revenue and increasing their probability of default. This is perfectly illustrates WWR: the counterparty owes you the most at the very moment when they are least likely to be able to pay. From this example, we can see that the main concern is foreign exchange WWR instead of interest rate swap WWR.

2.1.1 Forward Contract

A forward contract represents a binding obligation for the holder to purchase the underlying asset S at the agreed-upon delivery price K on the expiry date T.

Long Forward Position A long forward contract is an obligation for the holder to purchase the underlying asset.

- **Payoff at Expiry ($t = T$):**

$$V(S_T, T) = S_T - K \tag{1}$$

- **Value at time $t < T$:**

$$V(S_t, t) = S_t e^{-q(T-t)} - K e^{-r(T-t)} \tag{2}$$

where:

- S_t is the spot price of the underlying asset at time t,
- K is the strike price,
- T is the time to maturity,
- r is the continuously compounded risk-free interest rate,
- q is the continuously compounded dividend yield of the underlying asset,

Short Forward Position A short forward contract is an obligation for the holder to **sell** the underlying asset. This creates an inverted risk exposure relative to the long position.

- **Payoff at Expiry ($t = T$):**

$$V_(S_T, T) = K - S_T \tag{3}$$

- **Value at time $t < T$:**

$$V(S_t, t) = K e^{-r(T-t)} - S_t e^{-q(T-t)} \tag{4}$$

2.1.2 European Call Option

A European call option provides the holder with the right, but not the obligation, to purchase the underlying asset S at a predetermined strike price K on the expiry date T. Its value is derived from the potential for the asset price to rise. Let S_t be the underlying asset price at time t,

- **Payoff at Expiry ($t = T$):**

$$V_{\text{call}}(S_T, T) = \max(S_T - K, 0) \tag{5}$$

- **Value at time $t < T$ (Black-Scholes-Merton Formula):**

$$V_{\text{call}}(S_t, t) = S_t e^{-q(T-t)} \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2) \tag{6}$$

where:

- $\Phi(\cdot)$ is the standard normal cumulative distribution function,
- S_t is the spot price of the underlying asset at time t,
- K is the strike price,
- T is the time to maturity,
- r is the continuously compounded risk-free interest rate,
- q is the continuously compounded dividend yield of the underlying asset,
- σ is the implied volatility of the underlying asset's returns,

and the terms d_1 and d_2 are given by:

$$d_1 = \frac{\ln(S_t/K) + (r - q + \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T - t}}$$
$$d_2 = d_1 - \sigma\sqrt{T - t}$$

2.1.3 European Put Option

A European put option provides the holder with the right, but not the obligation, to sell the underlying asset S at a predetermined strike price K on the expiry date T. Its value is derived from the potential for the asset price to fall, creating an inverted exposure profile relative to a call option.

- **Payoff at Expiry ($t = T$):**

$$V_{\text{put}}(S_T, T) = \max(K - S_T, 0) \tag{7}$$

- **Value at time $t < T$ (Black-Scholes-Merton Formula):**

$$V_{\text{put}}(S_t, t) = Ke^{-r(T-t)}\Phi(-d_2) - S_t e^{-q(T-t)}\Phi(-d_1) \tag{8}$$

2.2 Stochastic Models for Portfolio Increments

The value and risk profile of the European derivatives discussed in the previous section are inherently dependent on the assumed stochastic process S_t, which governs the movement of their underlying asset prices. To ensure the robustness and comprehensiveness of the analysis, this dissertation employs three different models, including probability models and stochastic processes. These models have been selected as they act as the representatives of broader classes of models that capture different market dynamics. These models have been carefully selected to present complexity of a real world market, moving from a fundamental benchmark to an advanced framework that captures key empirical features of financial markets. This comparative approach allows us to analyse different market assumptions, regarding asset price volatility, such as constant volatility, stochastic volatility, and the presence of market jumps.

2.2.1 Geometric Brownian Motion (GBM)

The foundational model for Black-Scholes pricing, GBM, assumes the asset follows the stochastic differential equation (SDE):

$$dS_t = (r - q)S_t dt + \sigma S_t dW_t \tag{9}$$

where S_t is the price of the underlying asset at time t, r is the risk-free rate, q is the dividend yield, σ is the constant volatility, and W_t is a Brownian Motion.

Discretisation Scheme The SDE for GBM has an exact analytical solution. We can start solving the SDE by applying Itô's Lemma to $\log(S_t)$:

$$d(\log S_t) = \left(r - q - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t \tag{10}$$

$$\begin{aligned}
\ln(S_{t_{i+1}}) - \ln(S_i) &= \int_{t_i}^{t_{i+1}} \left(r - q - \frac{1}{2}\sigma^2\right) du + \int_{t_i}^{t_{i+1}} \sigma dW_u \\
&= \left(r - q - \frac{1}{2}\sigma^2\right) \Delta t_i + \sigma(W_{t_{i+1}} - W_{t_i})
\end{aligned}$$

The increment of a Brownian Motion, $W_{t_{i+1}} - W_{t_i}$, is a normally distributed random variable with mean 0 and variance Δt_i. We can therefore write it as $Z\sqrt{\Delta t_i}$, where $Z \sim \mathcal{N}(0, 1)$ and $\Delta t_i = t_{i+1} - t_i$. Substituting this in gives:

$$\ln\left(\frac{S_{t_{i+1}}}{S_{t_i}}\right) = \left(r - q - \frac{1}{2}\sigma^2\right) \Delta t_i + \sigma\sqrt{\Delta t_i} Z$$

Exponentiating both sides yields the formula for $S_{t_{i+1}}$:

$$S_{t_{i+1}} = S_{t_i} \exp\left((r - q - \frac{1}{2}\sigma^2)\Delta t_i + \sigma\sqrt{\Delta t_i} Z\right) \tag{11}$$

2.2.2 Heston Stochastic Volatility

The Heston model addresses one of the most significant empirical flaws of the Black-Scholes-Merton model: the assumption of constant volatility. The Black-Scholes-Merton framework, which is based on the GBM, predicts that the implied volatility of an asset option should be constant across all strike prices. However, the implied volatility observed in the real world exhibits a smile-like curve, subsequently coined the term "volatility smile", which the Heston model can reproduce.

Furthermore, the Heston model's correlation parameter, ρ, allows it to capture important empirical features of different asset classes. A crucial example, particularly relevant to equity markets, is the leverage effect, defined as the observed tendency for volatility to increase as asset prices decline. This phenomenon is modelled by setting the correlation parameter to a negative value ($\rho < 0$), linking adverse market movements to an increase in market stress. Conversely, in other markets, such as commodities or foreign exchange, the correlation can be positive ($\rho > 0$), where rising prices are often associated with increased volatility. This characteristic is crucial for realistically simulating market dynamics during stress events, making the Heston model a powerful extension of the benchmark GBM model shown in Section 2.2.1.

In this model, volatility is modelled as a stochastic, mean reverting process. The risk-neutral dynamics are described by the following system of SDEs:

$$dS_t = (r - q)S_t dt + \sigma S_t \sqrt{z_t} dW_t^{(1)} \tag{12}$$

$$dz_t = \kappa(1 - z_t)dt + \eta\sqrt{z_t} dW_t^{(2)} \tag{13}$$

where r is the risk-free interest rate, q is the dividend yield, S_t is the price of the underlying asset at time t, z_t is a stochastic variance multiplier, κ is the speed of its mean reversion to the long-term average mean is 1, and σ is a scaling parameter. The Brownian motions are correlated such that $dW_t^{(1)} dW_t^{(2)} = \rho_{Heston} dt$. Note that z_t is not the variance; the variance is given by the product $\sigma^2 z_t$.

η is formally known as the volatility of the variance, often referred to by the industry term 'vol of vol'. It determines the magnitude of the random shocks in the stochastic variance multiplier, z_t. While the mean-reversion term, $\kappa(1-z_t)dt$, deterministically pulls the variance towards its long-term average, the stochastic term, $\eta\sqrt{z_t} dW_t^{(2)}$, introduces randomness.

A higher η value indicates that the variance process itself exhibits greater volatility, subject to larger and more frequent shocks. This parameter is crucial for capturing the phenomenon of volatility clustering – a well-documented empirical feature of the financial markets, where periods of high volatility often follow periods of even higher volatility, while periods of calm tend to persist. By allowing for randomness in variance, the Heston model can more faithfully replicate market dynamics compared to the GBM model's assumption of constant volatility.

Discretisation Scheme The Heston model does not have a simple closed-form solution and must be discretised numerically. To discretise the Heston SDE, a log-Euler scheme is used for the asset price S_t to ensure positivity, while an Euler-Maruyama scheme with a full truncation method is applied to the variance process z_t. We will first introduce the Euler-Maruyama scheme.

Theorem (Euler-Maruyama Method). *The Euler-Maruyama method is a fundamental technique for obtaining a numerical solution to a general Itô stochastic differential equation (SDE) of the form:*

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t \tag{14}$$

where $a(X_t, t)$ is the drift term, $b(X_t, t)$ is the diffusion term, and dW_t is a Brownian Motion.

The method approximates the continuous SDE by a discrete-time equation over small time steps of size $\Delta t_i = t_{i+1} - t_i$. The differential dt is replaced by Δt_i, and the increment of the Wiener process, dW_{t_i}, is approximated by a random sample from a normal distribution with mean 0 and variance Δt_i. This leads to the following discrete update rule:

$$X_{t_{i+1}} \approx X_{t_i} + a(X_i, t_i)\Delta t_i + b(X_i, t_i) Z_i \sqrt{\Delta t_i} \tag{15}$$

where Z_i is a standard normal random variable, $Z_i \sim \mathcal{N}(0, 1)$. [10]

We can directly apply the Euler-Maruyama method to equation (13). Noting that $a(z_t, t) = \kappa(1 - z_t)$ and $b(z_t, t) = \eta\sqrt{z_t}$, we can write down the discrete update rules for z_i over a time step Δt_i.

$$z_{t_{i+1}} \approx z_{t_i} + \kappa(1 - z_i)\Delta t_i + \eta\sqrt{z_i}\Delta W_{t_i}^{(2)} \tag{16}$$

Similarly, we can also apply the same theorem to equation (12). However, there is a potential problem: S_t could potentially be negative, which is clearly not realistic, as $S_t = 0$ implies that the stock will be de-listed, thus it would never be negative. To tackle this issue, we consider the SDE for $\log(S_t)$ via Itô's lemma.

$$d(\log S_t) = \left(r - q - \frac{1}{2}\sigma^2 z_t\right) dt + \sigma\sqrt{z_i} dW_t^{(1)} \tag{17}$$

Now we can apply the Euler-Maruyama scheme to $\log(S_t)$, noting that $a(\log S_t, t) = r - q - \frac{1}{2}\sigma^2 z_t$ and $b(\log S_t, t) = \sigma\sqrt{z_t}$,

$$\log S_{t_{i+1}} - \log S_{t_i} \approx (r - q - \frac{1}{2}\sigma^2 z_i)\Delta t_i + \sigma\sqrt{z_i}\Delta W_{t_i}^{(1)} \tag{18}$$

Exponentiating both sides yield

$$S_{t_{i+1}} \approx S_{t_i} \exp\left((r - q - \frac{1}{2}\sigma^2 z_i)\Delta t_i + \sigma\sqrt{z_i}\Delta W_{t_i}^{(1)}\right) \tag{19}$$

Equation (16) and (19) gives us an initial look of its respective discretisation schemes. However, on a closer look, both of the equations have the term $\Delta W_t^{(k)}$, and the Brownian motions $W_t^{(1)}$ and $W_t^{(2)}$ are correlated via $dW_t^{(1)} dW_t^{(2)} = \rho_{Heston} dt$. This implies that we need to generate two independent variables, X_1 and X_2, such that they are both standard normal variables $\mathcal{N}(0, 1)$ with the relation $\mathbb{E}(X_1 X_2) = \rho_{Heston}$. Let Z_1 and Z_2 be two independent standard normal variables. Assume:

1. $X_1 = Z_1$

2. X_2 is a linear combination of Z_1 and Z_2 such that it is also a standard normal variable with correlation ρ_{Heston} with X_1:

$$X_2 = aX_1 + bX_2 \tag{20}$$

   where $a, b \in \mathbb{R}$.

We can solve for constants a and b via the following two conditions:

$$\begin{cases} \mathbb{E}[X_1 X_2] = \mathbb{E}[Z_1(aZ_1 + bZ_2)] & \text{(21a)} \\ \mathrm{Var}(X_2) = 1 & \text{(21b)} \end{cases}$$

From equation (21a),

$$\begin{aligned} \mathbb{E}[X_1 X_2] &= \mathbb{E}[Z_1(aZ_1 + bZ_2)] \\ \rho_{Heston} &= a\mathbb{E}[Z_1^2] + b\mathbb{E}[Z_1 Z_2] \end{aligned}$$

As $\mathbb{E}(Z_1^2) = \mathrm{Var}(Z_1) = 1$ and $\mathbb{E}(Z_1 Z_2) = 0$ as Z_1, Z_2 are uncorrelated, we obtain $a = \rho_{Heston}$.

From equation (21b),

$$\begin{aligned}
\mathrm{Var}(W_2) &= \mathrm{Var}(aZ_1 + bZ_2) \\
1 &= a^2\mathrm{Var}(Z_1) + b^2\mathrm{Var}(Z_2) \quad \text{(since } Z_1, Z_2 \text{ are independent)} \\
1 &= a^2 + b^2
\end{aligned}$$

Therefore $b = \sqrt{1 - \rho_{Heston}^2}$.
Substituting a and b into (20), we obtain the expression for W_2:

$$X_2 = \rho_{Heston} Z_1 + \sqrt{1 - \rho_{Heston}^2} Z_2$$

Finally, to obtain $\Delta W_{t_i}^{(1)}$ and $\Delta W_{t_i}^{(2)}$, which has a variance of Δt_i, we simply scale our constructed standard normal variables by $\sqrt{\Delta t_i}$:

$$\Delta W_{t_i}^{(1)} = \sqrt{\Delta t_i} W_1 = \sqrt{\Delta t_i} Z_1 \tag{22}$$

$$\Delta W_{t_i}^{(2)} = \sqrt{\Delta t_i} W_2 = \sqrt{\Delta t_i} (\rho_{Heston} Z_1 + \sqrt{1 - \rho_{Heston}^2} Z_2) \tag{23}$$

Substituting these results into (16) and (19) gives

$$\begin{cases}
z_{t_{i+1}} \approx z_{t_i} + \kappa(1 - z_i)\Delta t_i + \eta\sqrt{z_i \Delta t_i}(\rho_{Heston} Z_1 + \sqrt{1 - \rho_{Heston}^2} Z_2) & \text{(24a)} \\
S_{t_{i+1}} \approx t_i \exp\left((r - q - \frac{1}{2}\sigma^2 z_i)\Delta t_i + \sigma\sqrt{z_i \Delta t_i} Z_1\right) & \text{(24b)}
\end{cases}$$

A significant practical issue arises with the Euler-Maruyama scheme for the positive variance process. The stochastic term, $\eta\sqrt{z_i \Delta t_i}(\rho_{Heston} Z_1 + \sqrt{1 - \rho_{Heston}^2} Z_2)$, is a normally distributed random variable with a mean of zero. Therefore, it is possible for this term to take on a large negative value that is greater in magnitude than the positive drift and current variance, which would cause the next simulated variance, z_{i+1}, to become negative. A negative variance is financially and mathematically meaningless, and its square root is undefined, which would cause the simulation to fail.

The same problem can be also be found in the original SDE in equation (13). This is illustrated by the Feller condition, given by $2\kappa \geq \eta^2$ (with the long term mean of variance normalised to 1). This determines whether the variance process z_t can ever reach 0. If the condition is met, then the mean reverting drift $\kappa(1 - z_t)dt$ is sufficiently large such that z_t never reaches zero (see [7]).

However, Feller's condition is often violated when we are working under real-world parameters. This implies that, within realistic market scenarios, the magnitude of random shocks governed by the "volatility of variance" η is sufficient to drive the variance process towards zero. This situation precisely corresponds to the scenario where the standard Euler-Maruyama scheme is known to be numerically unstable, as it is prone to generating negative variance values and causing simulation breakdown. Furthermore, even if the Feller conditions are satisfied, the Euler-Maruyama scheme could still produce negative z_t. This is because the expression of z_t contains standard normal distributions Z_1 and Z_2, and over a finite time step Δt_i, Z_1 and Z_2 could take on large negative values, which might lead to a negative $z_{t_{i+1}}$. Now that we have justified the necessity of introducing specific modifications to the Euler-Maruyama method in order to ensure the positivity of the variance process z_t, we will introduce a method that satisfies our conditions called the Full Truncation Scheme (see [11]) is formally stated below.

Beyond these features, a key reason for using the Heston model is its computational efficiency. While it lacks a simple closed form solution like the Black-Scholes-Merton formula, it possesses a semi-analytical solution for European options based on Fourier transforms. This allows for the practical and quick calculation of option prices in the real world, which is essential for the efficient calibration of the model's parameters such as volatility.

Theorem (Full Truncation Scheme)**.** *The Full Truncation scheme modifies the standard Euler-Maruyama update rule for the Heston variance process by replacing the discretised variance term z_{t_i} within the stochastic component with its positive part, $z_{t_i}^{\max} = \max(z_{t_i}, 0)$. The update rule is given by:*

$$z_{t_{i+1}} \approx z_{t_i} + \kappa(1 - z_{t_i})\Delta t_i + \eta\sqrt{z_{t_i}^{\max}\Delta t_i}(\rho_{Heston}Z_1 + \sqrt{1 - \rho_{Heston}^2}Z_2) \tag{25}$$

This same truncated variance, $z_{t_i}^{\max}$, is then used in the discretization of the log-price SDE to maintain consistency and ensure the stability of the entire system. [11]

An alternative to truncation is the Reflection scheme (see [4] & [11]), where the variance process z_{t_i} is replaced by its absolute value, $|z_{t_i}|$. While this scheme can have better convergence properties for certain parameter sets, it can also lead to numerical instability, particularly when the volatility of variance, η, is high. Given that this analysis is focused on stress events where high volatility is a key feature, the guaranteed stability of the Full Truncation scheme is a more attractive option.

By replacing z_{t_i} with $z_{t_i}^{\max}$ in the diffusion term of the variance SDE and in both the drift and diffusion terms of the log-price SDE, we ensure that the simulation remains numerically stable and that the generated paths are economically sensible, with both stock prices and their variances guaranteed to be non-negative. We finally arrive at the discretisation scheme:

$$\begin{cases} z_{t_{i+1}} \approx z_{t_i} + \kappa(1 - z_{t_i})\Delta t_i + \eta\sqrt{z_{t_i}^{\max}\Delta t_i}(\rho_{Heston} Z_1 + \sqrt{1 - \rho_{Heston}^2} Z_2) & \text{(26a)} \\ S_{t_{i+1}} \approx S_{t_i} \exp\left((r - q - \frac{1}{2}\sigma^2 z_{t_i}^{\max})\Delta t_i + \sigma\sqrt{z_{t_i}^{\max}\Delta t_i} Z_1\right) & \text{(26b)} \end{cases}$$

2.2.3 Merton Jump-Diffusion

The Merton model addresses another limitation of the GBM model: the assumption of price path continuity. Real-world asset prices are known to exhibit sudden, discontinuous "jumps", particularly during periods of market stress or when significant news events occur. Such jumps result in fat-tailed return distributions, a phenomenon not captured by some continuous diffusion models such as the GBM model.

The Merton model was selected because it provides the most direct and practical method to incorporate this "gap risk" into the analysis. This choice is particularly relevant within the context of this dissertation, as the event-driven WWR framework proposed by Dickinson [8] is specifically designed to model default events triggered by precisely this type of market gap. While this dissertation adopts Pykhtin's statistical framework, incorporating the Merton model enables the verification of its performance within the core market dynamics driving alternative WWR models, thereby providing a crucial benchmark for comparison.

To capture the potential for sudden, large price movements (fat tails), this model adds a compound Poisson process q_t to the GBM model. The risk-neutral process for the stock price follows the SDE:

$$\frac{dS_t}{S_t} = (r - q - \lambda\bar{k})dt + \sigma dW_t + k dq_t \tag{27}$$

where r is the risk-free interest rate, q is the dividend yield, σ is the volatility, q_t is a Poisson process with intensity λ, representing the arrival of jumps. The term k is a random variable representing the percentage jump size, whose distribution obeys $\log(1 + k) \sim \mathcal{N}(\gamma, \delta^2)$, and $\bar{k} = e^{\gamma+\delta^2/2} - 1$ is its mean. The term $(r - q - \lambda\bar{k})$ is the risk-neutral drift, which is compensated for the expected return from the jumps.

A key feature of the Merton model is its ability to generate the fat-tailed distributions observed in financial markets. This stems from the model's structure, which combines a continuous diffusion process with a compound Poisson process. The diffusion component models typical, small market fluctuations, while the jump component introduces the possibility of rare, large, discontinuous price movements—events of low probability but high impact. It is this combination that populates the tails of the return distribution. To show how the fat tails are produced, we introduce the probability density of the log-return $x_t = \log(S_t/S_0)$ (modified from [12], Equation 7):

$$\mathbb{P}(x_t) = \sum_{i=0}^{\infty} \frac{e^{-\lambda t}(\lambda t)^i}{i!} N\Big(x_t;\, (\alpha - \tfrac{\sigma^2}{2} - \lambda k)t + i\gamma,\, \sigma^2 t + i\delta^2\Big) \tag{28}$$

The fat tails arise because each individual jump has mean γ and variance δ^2, so if there are i jumps then it contributes to a fixed shift of $i\gamma$ and fixed variance $i\delta^2$ that does not depend on time t. However, the probability of i jumps occurring is governed by:

$$\mathbb{P}(i \text{ jumps}) = \frac{e^{-\lambda t}(\lambda t)^i}{i!} \tag{29}$$

which decays exponentially with λt. Hence when the number of jump i increases, the probability of a jump occurring exponentially decreases, but the corresponding normal distribution (in Equation (28)) has a large variance $\sigma^2 t + i\delta^2$ and a shifted mean $i\gamma$. This distribution with low probability with high variance exactly creates the fat tails characteristic.

In fact, the total variance of the asset's log-return over a period t is the sum of the variance from the continuous diffusion and the variance from the jumps:

$$\mathrm{Var}(\log(S_t/S_0)) = \sigma^2 t + \lambda(\gamma^2 + \delta^2)t \tag{30}$$

Discretisation Scheme The Merton model SDE has an exact analytical solution. In a process with discontinuities, a jump at time t causes an instantaneous change in the asset price. We denote the price immediately before the jump as S_{t-}, and the price immediately after the jump as S_t. Now we can apply Itô's Lemma with jumps to the process $Y_t = \log(S_t)$. For a jump process, the lemma states that the change in the function due to a jump is $f(S_t) - f(S_{t-}) = \log(S_t) - \log(S_{t-}) = \log(S_t/S_{t-})$. Since the percentage jump size is $k = S_t/S_{t-} - 1$, we have $S_t/S_{t-} = 1 + k$, and the change in the log-price during a jump is $\log(1 + k)$.

Applying the full Itô's Lemma for jump-diffusion processes gives the SDE for $\log(S_t)$. To show this derivation clearly, we first formally state Itô's lemma with for jump diffusion processes.

Theorem (Itô's Lemma for Jump-Diffusion Processes [6])**.** *Suppose a stochastic process X_t follows the SDE:*

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t + J(X_{t-}, t)dq_t$$

where dq_t is the differential of a Poisson process. For a twice-differentiable function $f(X_t, t)$, its differential df is given by:

$$df = \left(\frac{\partial f}{\partial t} + a\frac{\partial f}{\partial x} + \frac{1}{2}b^2\frac{\partial^2 f}{\partial x^2}\right)dt + b\frac{\partial f}{\partial x}dW_t + [f(X_{t-} + J) - f(X_{t-})]dq_t$$

We apply this lemma to our function $Y_t = \log(S_t)$, where S_t follows Equation (27). First, we identify the components:

- Function: $f(S_t) = \log(S_t)$
- Drift: $a = (r - q - \lambda\bar{k})S_t$
- Diffusion: $b = \sigma S_t$
- Jump Size: $J = kS_{t-}$

Note that $\frac{\partial f}{\partial s} = \frac{1}{S_t}$, $\frac{\partial^2 f}{\partial s^2} = -\frac{1}{S_t^2}$ and $[f(S_{t-} + kS_{t-}) - f(S_{t-})]dq_t = [\log(S_{t-}(1 + k)) - \log(S_{t-})]dq_t = \log(1 + k)dq_t$. Substituting these into the lemma gives the SDE for $\log S_t$:

$$d(\log S_t) = \left(r - q - \lambda\bar{k} - \frac{1}{2}\sigma^2\right)dt + \sigma dW_t + \log(1 + k)dq_t \tag{31}$$

This process is composed of three independent components: a deterministic drift, a continuous diffusion, and a jump process. We can integrate this SDE directly over a discrete time interval $\Delta t_i = t_{i+1} - t_i$:

$$\int_{t_i}^{t_{i+1}} d(\log S_u) = \int_{t_i}^{t_{i+1}} \left(r - q - \lambda\bar{k} - \frac{1}{2}\sigma^2\right)du + \int_{t_i}^{t_{i+1}} \sigma dW_u + \int_{t_i}^{t_{i+1}} \log(1 + k)dq_u$$
$$\log(S_{t_{i+1}}) - \log(S_i) = \left(r - q - \lambda\bar{k} - \frac{1}{2}\sigma^2\right)\Delta t_i + \sigma(W_{t_{i+1}} - W_{t_i}) + \sum_{j=1}^{N_i}\log(1 + k_j)$$

where N_i is the number of jumps that occur in the interval Δt_i, and k_j are the percentage sizes of those jumps.

To simulate this, we use the following facts:

- The continuous diffusion increment, $\sigma(W_{t_{i+1}} - W_{t_i})$, is simulated as $\sigma\sqrt{\Delta t_i}Z$, where $Z \sim \mathcal{N}(0, 1)$.
- The number of jumps, N_i, is a random variable drawn from a Poisson distribution, $N_i \sim \text{Poi}(\lambda\Delta t_i)$.

- For each jump j, the size k_j is drawn from the specified distribution where $\log(1+k_j) \sim \mathcal{N}(\gamma, \delta^2)$.

This gives the update rule for the log-price:

$$\log\left(\frac{S_{t_{i+1}}}{S_i}\right) = \left(r - q - \lambda\bar{k} - \frac{1}{2}\sigma^2\right)\Delta t_i + \sigma\sqrt{\Delta t_i}Z + \sum_{j=1}^{N_i}\log(1+k_j)$$

Exponentiating both sides yields the final simulation formula. The sum of logarithms in the jump term becomes a product after exponentiation: $\exp(\sum\log(1+k_j)) = \prod\exp(\log(1+k_j)) = \prod(1+k_j)$. This brings us to the final result:

$$S_{t_{i+1}} = S_i \exp\left((r - q - \lambda\bar{k} - \frac{1}{2}\sigma^2)\Delta t_i + \sigma\sqrt{\Delta t_i}Z\right)\prod_{j=1}^{N_i}(1+k_j) \tag{32}$$

Simulation of Jumps Recall that the number of jumps N_i is distributed by $\mathrm{Poi}(\lambda\Delta t_i)$, where the probability of n jumps in a time step Δt_i is:

$$P(N_i = n) = \frac{(\lambda\Delta t_i)^n e^{-\lambda\Delta t_i}}{n!} \tag{33}$$

Hence the probability of more than 1 jumps in time steps Δt_i is:

$$\begin{aligned}
P(N_i > 1) &= 1 - P(N_i = 0) - P(N_i = 1) && (34)\\
&= 1 - e^{-\lambda\Delta t_i} - \lambda\Delta t_i e^{-\lambda\Delta t_i} && (35)\\
&= 1 - (1 - \lambda\Delta t_i + \frac{1}{2}(\lambda\Delta t_i)^2) - \lambda\Delta t_i(1 - \lambda\Delta t_i) + \mathcal{O}((\lambda\Delta t_i)^3) && (36)\\
&\approx \frac{1}{2}(\lambda\Delta t_i)^2 && (37)
\end{aligned}$$

where we used the 1st order and 2nd order Taylor Series Expansion on e^{-x} and xe^{-x} respectively in Equation (36).

If we take small time steps Δt_i used in this analysis (such as the 10-day Margin Period of Risk, where $\Delta t_i \approx 10/252$), then the probability of more than one jump occurring in the time step Δt_i is small, so we make a key assumption: the probability of more than one jump occurring in a single interval Δt_i is negligible. This allows the Poisson process to be approximated by a Bernoulli trial for each time step Δt_i. If a jump is triggered, its size, k, is then drawn from the log-normal distribution, captured by $\log(1+k) \sim \mathcal{N}(\gamma, \delta^2)$. The simulation algorithm implemented is therefore as follows:

1. For each time step from t_i to t_{i+1}, generate a single random number, u, from a uniform distribution on [0, 1].

2. If $u < P(N_i \geq 1) = 1 - P(N_i = 0) = 1 - e^{-\lambda \Delta t_i} \approx \lambda \Delta t_i$, a single jump is deemed to have occurred within the period Δt_i (i.e., $N_i = 1$). The approximation in the equation uses the 1st order Taylor Series expansion on e^{-x}.

3. Otherwise, no jump occurs ($N_i = 0$).

3 Modelling CVA Framework with Wrong Way Risk

3.1 Foundational CVA Formulation

The Credit Valuation Adjustment (CVA) represents the market price of counterparty credit risk. The CVA is formally defined as the risk-neutral expectation of the discounted loss, given by:

$$\text{CVA} = (1 - R) \int_0^T EE^*(t) d\text{PD}(0, t) \tag{38}$$

where R is the recovery rate, $(1-R)$ is the loss given default, and $d\text{PD}(0, t)$ is the risk-neutral probability of the counterparty defaulting between time 0 and t. The term $\text{EE}^*(t)$ is the risk-neutral discounted expected exposure at time t, defined as:

$$EE^*(t) = \mathbb{E}^{\mathbb{Q}} \left[\frac{B_0}{B_t} EE(t) \right] \tag{39}$$

Here, $B_t = e^{rt}$ is the value of the risk-free money market account at time t, which acts as the numéraire for discounting. The term $EE(t)$ represents the undiscounted credit exposure at time t. Suppose that a financial institution which has a portfolio of derivatives traded with a counterparty. Then the exposure $EE(t)$ of the financial institution to the counterparty is given by [1]:

$$EE(t) = \max(V(t) - V(t - \delta) - IM(t), 0) \tag{40}$$

where $V(t)$ is the value of portfolio at time t and $IM(t)$ is the value of the initial margin at time t. In this case, we are assuming that all cash flows are paid by the counterparty and the financial institution for the entire period of MPoR.

It is important to note that the expected exposure $EE^*(t)$ is taken under the risk neutral measure $\mathbb{Q}$, which ensures that the CVA is an arbitrage-free price for counterparty risk.

The fundamental component of this formula is the model for the counterparty's probability of default. This analysis adopts the deterministic hazard rate model. This approach assumes that the default event can be modelled as the first jump in a Poisson process. To simplify the treatment, we begin by considering the simplest scenario of a constant hazard rate λ. This implies that the cumulative default probability up to time t is given by the equation:

$$\text{PD}(0, t) = 1 - e^{-\lambda t} \tag{41}$$

It is important to acknowledge that this is a simplification. In practice, it is more common to use a time-dependent deterministic hazard rate, which provides the flexibility to match an entire curve of market-observed credit spreads. However, this additional complexity is excluded because it does not add significant value to the specific analysis in this dissertation.

The primary motivation for adopting a constant hazard rate model lies in its analytical tractability and its established status as the standard foundational model for credit risk analysis. This model operates on the "memoryless" basis, providing a clear and intuitive baseline for default probabilities. This core assumption asserts that the likelihood of default remains constant over time is unaffected by the counterparty's past record. Adopting this constant hazard rate assumption significantly simplifies the analytical framework, establishing an ideal and easily interpretable foundation for analysing the impact of WWR.

However, the major limitation of the constant hazard models lies in the assumption that default events and market factors are mutually independent. Constant hazard models assumes that default events are independent of all other sources, including events such as market shocks. The collapse of the Archegos case demonstrated a fundamental flaw in this assumption – at that time, market shocks and default events were closely intertwined. This necessitates the introduction of a more precise framework that explicitly links market movements to default probabilities. To address this challenge, Dickinson and Pythkin proposed two main approaches. In Mind The Gap [8], Dickinson modelled default events as being directly triggered by a sudden jump in prices, using Lévy processes. On the other hand, in Weighting For Leverage [14], Pykhtin adopted a distribution-based approach in which a default is not linked to a jump but rather to the change in portfolio value. His model uses a Gaussian Copula (see Section 3.2) to correlate the default event with the portfolio's value change.

Although Dickinson's event-driven framework provides a model for defaults arising from market gaps that aligns with physical intuition, this paper selected the Pykhtin's model based on key practical considerations. Pykhtin's model can be applied to the distribution of portfolio value changes under different deterministic and stochastic processes generated by Monte Carlo simulations, such as the GBM, Heston, and Merton models (see Section 2.2). In contrast, adopting Dickinson's approach based on the Lévy process necessitates constructing an entirely new, non-standard simulation engine, making comparative analysis beyond the underlying dynamics difficult. Adopting the Pythkin framework enables robust verification of the core risk premium assumptions inherent in various standard models, under conditions that eliminate the influence of customised simulation methodologies.

Now we can work on discretising the continuous integral CVA over a series of time steps t_i:

$$\text{CVA} \approx (1 - R) \sum_{i=1}^{N} \text{EE}^*(t_i) \cdot \Delta\text{PD}(t_i) \tag{42}$$

The discretised expected exposure $\text{EE}^*(t_i)$ is given by $e^{-rt_i}\text{EE}(t_i)$. The probability of default is modelled using a constant hazard rate, λ, such that the cumulative probability of defaulting by time t is $\text{PD}(0, t) = 1 - e^{-\lambda t}$. The incremental default probability over a period $[t_{i-1}, t_i]$ is therefore $\Delta\text{PD}(t_i) = e^{-\lambda t_{i-1}} - e^{-\lambda t_i}$. This gives us

$$\text{CVA} \approx (1 - R) \sum_{i=1}^{N} e^{-rt_i}\text{EE}(t_i) \cdot (e^{-\lambda t_{i-1}} - e^{-\lambda t_i}) \tag{43}$$

The main difficulty in finding the CVA lies in accurately modelling the expected exposure which becomes much more complex when wrong-way risk is incorporated.

3.2 Modelling Leveraged Wrong Way Risk

The Archegos default demonstrated that conventional models, which often assume independence between market movements and a counterparty's default, are insufficient for highly leveraged entities. To address this, we adopt the leveraged WWR framework proposed by Pykhtin. His framework explicitly links the counterparty's default event to large, adverse increments in the portfolio's market value over the margin period of risk (MPoR).

The challenge of linking the default event and the continuous distribution of portfolio value changes requires tools capable of constructing joint probability distributions from their constituent elements. For this reason, we decide on using the Gaussian copula. A Gaussian copula is a mathematical tool used to describe the dependence between multiple random variables. Its primary function is to "combine" the individual marginal distributions of each variable through a predefined dependency structure, thereby constructing a joint probability distribution. The framework based on the copula was selected due to its powerful functionality, flexibility, and practical advantages. It enables the construction of the joint distribution of default events and portfolio value fluctuations using only their marginal distributions and a single correlation parameter p_{WWR}. This approach is remarkably concise and avoids restrictive assumptions about the underlying market dynamics. Crucially, this allows the impact of WWR to be investigated under a single, consistent framework across multiple stochastic processes for the underlying (GBM, Heston and Merton), whilst simultaneously separating market risk modelling from credit market dependency.

The model's core is a Gaussian copula that correlates the default event with the portfolio's value change, denoted by $\delta V(t-\delta, t)$. Here, we note that $\delta V(t-\delta, t)$ can be modelled by any of the 3 underlying models: GBM, Heston and Merton, using standard contract types (see Section 2). We can then map the distribution of δV to a standard normal variable, Z, via its empirical cumulative distribution function (CDF), F_t:

$$F_t(v) = \frac{1}{M}\sum_{m=1}^{M}\left(\mathbf{1}_{\{v>\delta V_\delta^{(m)}(t)\}} + \frac{1}{2}\mathbf{1}_{\{v=\delta V_\delta^{(m)}(t)\}}\right) \tag{44}$$

$$Z \equiv \Phi^{-1}(F_t[\delta V(t-\delta,t)]) \tag{45}$$

where Φ^{-1} is the inverse of the standard normal CDF and $\delta V(t-\delta,t) = V(t) - V(t-\delta)$, and $\delta V_\delta^{(m)}$ is the m^{th} realisation of the increment in portfolio value between time $t-\delta$ and t. This step effectively ranks the simulated outcomes of δV. A separate standard normal variable, X, governs the default event, which occurs if $X \leq \Phi^{-1}(h(t))$, where $h(t)$ is the forward incremental probability of default.

The link is established by correlating these two variables with a parameter $\rho \in [0,1]$:

$$X = -\rho Z + \sqrt{1-\rho^2}\epsilon \tag{46}$$

where ϵ is another independent standard normal variable. This structure ensures that as ρ_{WWR} increases, a large portfolio gain (high Z), which represents high exposure, becomes strongly correlated with the default event (low X).

The probability of default, denoted by $h^{(m)}(t)$, is conditional on a specific simulation path m (which is uniquely identified by its corresponding standard normal variable, $Z^{(m)}$). In particular, it is the probability that the default event occurs, given the outcome of the portfolio increment:

$$\begin{aligned} h^{(m)}(t) &\equiv P(\text{Default} \mid Z = Z^{(m)}) \\ &= P(X \leq \Phi^{-1}(h(t)) \mid Z = Z^{(m)}) \\ &= P(-\rho Z^{(m)} + \sqrt{1-\rho^2}\epsilon \leq \Phi^{-1}(h(t))) \end{aligned} \tag{47}$$

By rearranging the inequality to solve for the independent standard normal variable ϵ, we get:

$$h^{(m)}(t) = P\left(\epsilon \leq \frac{\Phi^{-1}(h(t)) + \rho Z^{(m)}}{\sqrt{1-\rho^2}}\right)$$

Since ϵ follows a standard normal distribution, this probability is given by the standard normal CDF, $\Phi(\cdot)$:

$$h^{(m)}(t) = \Phi\left(\frac{\Phi^{-1}(h(t)) + \rho Z^{(m)}}{\sqrt{1-\rho^2}}\right)$$

The weight for each path, $w^{(m)}(t)$, is defined as the ratio of this path-dependent default probability to the average, unconditional default probability, $h(t)$.

To arrive at the final formula, we replace the theoretical variable $Z^{(m)}$ with its numerical approximation from the simulation. Given M simulation paths sorted in increasing order of the portfolio increment, the empirical CDF for the m^{th} path is approximated as $(m-0.5)/M$. Therefore, $Z^{(m)}$ is given by:

$$Z^{(m)} \approx \Phi^{-1}\left(\frac{m-0.5}{M}\right)$$

Substituting this into the expression for $w^{(m)}(t) = h^{(m)}(t)/h(t)$ directly yields the weights of the m^{th} simulation path, $w^{(m)}(t)$, which systematically increases the importance of high-exposure paths and are given by:

$$w^{(m)}(t) = \frac{1}{h(t)}\Phi\left(\frac{\Phi^{-1}[h(t)] + \rho\Phi^{-1}[(m-0.5)/M]}{\sqrt{1-\rho^2}}\right) \tag{48}$$

where the simulation paths are sorted in increasing order of $\delta V^{(m)}$.

This leads to a WWR-adjusted Expected Exposure, calculated not as a simple average, but as a weighted average over M simulation paths:

$$\mathrm{EE}(t) = \frac{1}{M}\sum_{m=1}^{M} w^{(m)}(t)\left(\delta V^{(m)}(t) - \mathrm{IM}^{(m)}(t)\right)^{+} \tag{49}$$

where $\mathrm{IM}^{(m)}(t)$ is the mth realisation of $\mathrm{IM}(t)$. Initial Margin is the collateral posted at the beginning of the Margin Period of Risk, time t, designed to cover potential future losses over that period δ. For simplicity reason, we assume that $\mathrm{IM}(t)$ is deterministic and is determined as a Value at Risk (VaR) at some fixed confidence level, α. We can now define $\mathrm{IM}(t)$ under the significance level α:

$$\mathrm{IM(t)} = Q_{\delta V}(\alpha) \tag{50}$$

where $Q_{\delta V}(\alpha)$ is the quantile function of the empirical distribution of portfolio increments, $\{\delta V^{(m)}(t)\}_{m=1}^{M}$. The general quantile function, $Q_X(\alpha)$, is defined as:

$$Q_X(\alpha) = \inf\{x \in \mathbb{R} \mid F_X(x) \geq \alpha\} \tag{51}$$

While the general framework allows for a path-dependent initial margin $IM^{(m)}(t)$, it is important to note that the IM is a single value calculated from the distribution of simulated portfolio increments and is therefore not path-dependent. However, this is a reasonable assumption as this reflects how IM is applied and determined in the real world.

3.3 The CVA Calculation Procedure

The final CVA is calculated by integrating these components. For a given portfolio type and underlying model, the procedure is as follows:

1. Establish a time grid $\{t_1, \dots, t_N\}$ over the life of the transaction.

2. For each time step t_i:

   (a) Simulate M paths of the underlying asset from $t_i - \delta$ to t_i (where δ is the MPoR) using the chosen stochastic model (GBM, Merton, or Heston).

   (b) Using the chosen derivative pricing function (Call, Put, or Forward), calculate the initial value $V(t_i - \delta)$ and the final values $V(t_i)$ for each of the M paths. The difference constitutes the distribution of the portfolio increment, $\delta V^{(m)}(t_i)$.

   (c) Apply the WWR methodology from Section 2.2 to this δV distribution to calculate the WWR-adjusted $EE(t_i)$.

3. Integrate the resulting term structure of expected exposures, $\{\mathrm{EE}(t_i)\}$, against the discretised default probabilities, $\{\Delta\mathrm{PD}(t_i)\}$, using equation (43) to arrive at the final CVA value.

4 The Impact of WWR on Expected Exposure

Before performing a full CVA calculation, it is necessary to isolate and analyse the direct impact of the leveraged WWR model on expected exposure (EE). According to Pykhtin's framework, as the correlation parameter p increases, EE shift from being a simple average of risk exposures to a weighted average emphasising high-loss scenarios.

To quantify this effect, we replicate Pykhtin's [14] illustrative analysis, calculating how EE changes as a function of p_{WWR} for our 3 fundamental stochastic models δV shown in Section 2.3. The analysis is conducted for portfolios with no initial margin, 95% initial margin and 99% initial margin, with the results normalised by the standard deviation of the portfolio increment, δV.

The initial margin is also a significant factor in the expected exposure. The primary role of Initial Margin is to act as a substantial buffer against potential future losses. This mechanism is clear from Equation (49), in particular the term $(\delta V - \mathrm{IM})^+$. Any portfolio loss within the time interval $[t - \delta, t]$ that is smaller than the IM amount is fully absorbed by the collateral, resulting in zero credit exposure. Exposure only materializes for the tail of the distribution, specifically for those scenarios where losses exceed this pre-posted buffer.

This powerful risk-mitigating effect is immediately apparent in Figures 1 and 2. In the absence of wrong way risk ($\rho = 0$), the Expected Exposure for portfolios with 95% and 99% IM is drastically lower than for the uncollateralised (No IM) portfolio, starting at values very close to zero. This confirms that, in the absence of WWR, a high level of Initial Margin is extremely effective at covering the vast majority of potential future losses. Having established this baseline, the goal of this chapter is to quantify how this effectiveness diminishes as wrong-way risk is introduced (that is, when ρ increases). We will first explore the behaviour of an at-the-money call option where the underlying asset follows the Heston model.

As the results presented in Figure 1 show, this confirms that the expected exposure is an increasing and convex function of the WWR correlation. The analysis of the Geometric Brownian Motion model yields qualitatively similar results, indicating that the WWR effect is robust across continuous diffusion processes (see Appendix A).

To provide a contrast to the continuous diffusion models, the analysis was re-run using the Merton jump-diffusion model. The parameter settings aimed to simulate a situation where the primary tail risk in the market stems from large positive jump changes in asset prices ($\gamma = 0.25$). As shown in Figure 2, the results from the Merton model appears very different from those of the Heston model.

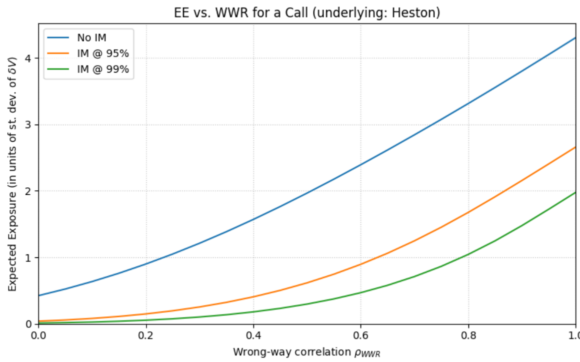

Figure 1: Expected Exposure against ρ, Heston Call model with $h(t) = 0.001$ and strike $K = 100$

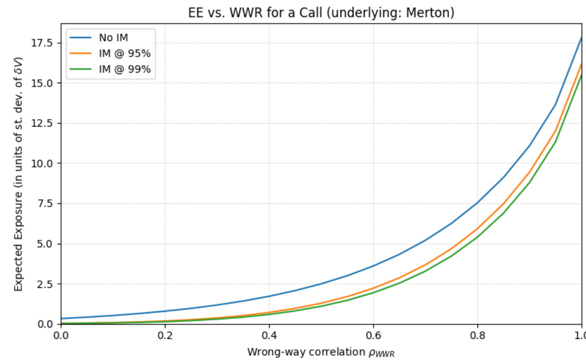

Figure 2: Expected Exposure against ρ, Merton Jump-Diffusion Call model with $h(t) = 0.001$and strike $K = 100$

For call options, rare yet abrupt upward jumps generate a "lottery effect", imparting a pronounced positive skew and kurtosis (i.e., right-tail heaviness) to the distribution of the portfolio's incremental δV. This heavy tail directly impacts standardized analysis in two respects. First, it induces an extremely large standard deviation in δV, causing a noticeable disparity in the order of magnitude on the y-axis when used as a normalisation factor. Second, and more critically, the initial margin becomes almost entirely dominated by jump risk, as it is calculated as a high percentile driven by jumps. This leads to substantially increased margin requirements. This elevated initial margin enables more effective coverage of extreme tail events amplified by the WWR model, explaining the widening gap between the guaranteed and non-guaranteed risk curves.

This comparative analysis demonstrates that the specific source of potential tail risk – whether stemming from continuous diffusion, random volatility, or discrete jumps – is a crucial factor in determining the interaction between leveraged WWR and collateral. This finding provides direct motivation for subsequent CVA calculations.

Table 1 quantifies the EE multiplier effect of WWR relative to the baseline case with zero correlation ($\rho = 0$). A key observation is that the impact of WWR is noticeably greater for portfolios with higher collateralisation ratios. For instance, at $\rho = 1$, the increase in EE for an uncollateralised portfolio reaches approximately 10-fold. In contrast, the amplification effect of extreme events in a 99% collateralised portfolio exceeds 170-fold. This phenomenon arises because the baseline extreme event effect in highly collateralised portfolios is extremely small, causing the increase driven by WWR to appear very large on a relative scale. This indicates that while collateral management effectively reduces conventional risk exposure, the residual tail risk reacts with extreme sensitivity to WWR.

This finding is further clarified in Table 2, which examines the risk-reducing effect of initial margin. The results reveal that the effectiveness of initial margin diminishes significantly as the WWR increases. At $\rho = 0$, a 99% initial margin level reduces the expected exposure by a remarkable 97.3% (from 1.6066 to 0.0431). However, at $\rho = 1$, the reduction is only 54.1% (calculated as $1-\mathrm{EE}(\rho > 0)/\mathrm{EE}(\rho = 0)$ based on Table 1). The fact that the protective effect is reduced represents a significant finding, suggesting that even high collateralisation ratios may not sufficiently guard against the correlated default risk arising from counterparties with extremely high leverage ratios. This detailed analysis of expected risk exposure directly motivates the subsequent full credit value adjustment (CVA) calculation.

Table 1: The multiplicative impact of leveraged WWR on Expected Exposure (Heston Model, At-the-Money Call Option). The table shows the baseline Expected Exposure at $\rho = 0$ and the ratio $\mathrm{EE}(\rho > 0)/\mathrm{EE}(\rho = 0)$ for various levels of WWR correlation.

		EE($\rho > 0$)/EE($\rho = 0$)				
IM Amount	**EE($\rho = 0$)**	**$\rho = 0.1$**	**$\rho = 0.25$**	**$\rho = 0.5$**	**$\rho = 0.75$**	**$\rho = 1.0$**
No IM	1.6066	1.5	2.5	4.6	7.3	10.2
IM @ 95%	0.1504	2.0	4.9	15.5	36.7	67.1
IM @ 99%	0.0431	2.3	6.6	26.1	76.1	173.9

Table 2: The effectiveness of Initial Margin in mitigating WWR (Heston Model, At-the-Money Call Option). The table shows the ratio of the Expected Exposure with IM to the Expected Exposure without IM for various levels of WWR correlation.

IM Amount	**$\rho = 0.0$**	**$\rho = 0.1$**	**$\rho = 0.25$**	**$\rho = 0.5$**	**$\rho = 0.75$**	**$\rho = 1.0$**
IM @ 95%	9.4%	12.7%	18.6%	31.3%	47.2%	61.8%
IM @ 99%	2.7%	4.1%	7.2%	15.1%	28.1%	45.9%

5 Deconstructing the Implicit WWR Risk Premium

The theoretical framework for quantifying leveraged wrong-way risk, as proposed by Pykhtin (2024), provides the foundation for the analysis in this chapter. The model, constructed and analysed in Chapter 2 and 3 of this dissertation, demonstrates that this effect exerts a significant influence on the CVA. The simulations conducted herein extended beyond simple Gaussian scenarios to include complex dynamics such as stochastic volatility and market jumps. The results indicate that, under a fixed baseline default probability, accounting for this adverse risk effect leads to a significant increase in expected exposure. This finding aligns with the direction of effects observed in Pykhtin's original analysis, which suggests that highly leveraged investment portfolios can generate amplification factors exceeding 300 times under $\rho = 1$ and IM at 99% confidence level. Even if we look at more realistic scenarios where the wrong way risk correlation $\rho = 0.5$, the amplification factor is still a significant 17 and 35 times under IM at 95% and 99% confidence level respectively. This result substantiates the core hypothesis – that conventional models may significantly underestimate the expected exposure, and consequently the CVA – while also establishing the starting point for the subsequent investigation: whether this theoretical amplification effect is as severe when evaluated using market-implied default probabilities.

However, a significant practical challenge arises when attempting to implement such theoretical WWR models. Within these frameworks, the benchmark default probability is treated as a given exogenous input variable. In practice, this default probability is typically derived from market instruments, most commonly the counterparty's credit default swap (CDS) spread. For example, the default probability implicitly signalled by the market is characteristically higher than historically observed default rates. This difference, known as the credit risk premium, functions as a market-wide buffer mechanism to compensate for various unmodelled risks – including the leveraged WWR that this dissertation seeks to quantify.

This creates a critical implementation question: should a risk manager simply layer an explicit WWR model (using $\rho > 0$) on top of these already elevated market parameters? Doing so risks "double-counting" the same risk, as the elevated market price of credit may already serve as an implicit compensation for the WWR that the model then seeks to add explicitly. In this chapter, we will address this crucial issue by shifting the analytical focus from merely quantifying the WWR effect to examining how this effect is potentially already embedded in the market's pricing of default risk.

This implementation challenge can be illustrated with an analogy. The market's pricing of credit risk, as reflected in the high hazard rate implied by CDS spreads, acts as a "blunt instrument." It bundles together pure default risk, liquidity risk, and an implicit premium for WWR into a single, observable price. In contrast, the Pykhtin model provides a "surgical tool" in the form of the correlation parameter, ρ, designed to precisely model the WWR component.

The motivation for the experiment in this chapter is therefore not to prescribe a specific method for combining these tools. Rather, it is to address a more fundamental economic question raised by the dramatic results of models like Pykhtin's. If the CVA underestimation is as severe as a naive application of the model suggests, it would imply that the true cost of trading with leveraged counterparties is economically prohibitive, potentially leading to a breakdown in market activity.

This dissertation hypothesises that the market is more efficient than Pykhtin's analysis, and that the "blunt" price of credit already contains a significant premium for WWR. To test our hypothesis, our experiment is built on the assumption that the CVA under no IM priced by the market (denoted by CVA*) is correct. We use this value as a constraint to deconstruct the market's "blunt" hazard rate (λ) into a lower hazard rate which we term the Equivalent Hazard Rate (λ^*). Our experiment aims to provide a more nuanced estimate of the true CVA for collateralised trades – one that shows that while the risk is real and the CVA is indeed underestimated by conventional models, the underestimation may not be so severe as to preclude rational economic activity and "doing business" at equilibrium.

5.1 Experimental Design and Methodology

To test this hypothesis, we design a quantitative experiment to disentangle the two primary risk components: the baseline probability of default and the structural WWR. The methodology is as follows:

1. **Establish a Baseline CVA (CVA*)**
   First, we establish a baseline for the total uncollateralised credit risk as priced by a simple market model. A baseline CVA, denoted by CVA*, is calculated for an uncollateralised (No IM) portfolio. This calculation assumes a complete absence of wrong-way risk ($\rho = 0$) and uses a baseline market-observed hazard rate, λ. This CVA* represents the total market price of the credit risk before any explicit WWR modelling or collateral is considered. We are assuming that this uncollateralised CVA is already correctly priced by the market.

2. **Calculate the Equivalent Hazard Rate (λ^*)**
   The core of the experiment is to re-allocate this total risk between the baseline default

probability and the explicit WWR model. For each level of WWR correlation ($\rho > 0$), we numerically solve for a new, lower hazard rate called the Equivalent Hazard Rate (λ^*). It is important to note that this calibration is specific to a single instrument at a given maturity; the resulting Equivalent Hazard Rate is not intended to be universal but serves to demonstrate the WWR premium effect for a specific. This is achieved using Brent's method, a robust root-finding algorithm.

Theorem (Brent's Method [5], [13])**.** *Brent's method [5] is a hybrid root-finding algorithm that combines the guaranteed convergence of the bisection method with the speed of faster, open methods like the Secant method and Inverse Quadratic Interpolation. The algorithm maintains a bracketing interval* $[a, b]$ *where the function* $f(x)$ *has opposite signs at the endpoints, guaranteeing a root exists within the interval.*

At each iteration, the algorithm chooses the most efficient step. When possible, its fastest and preferred step is Inverse Quadratic Interpolation. This method uses three prior points—let the current best guess for the root be b*, the previous best guess be* a*, and the third point be* c*—to fit an inverse quadratic function of the form* $x = f(y)$*. The next estimate for the root is found by evaluating this function at* $y = 0$*.*

The full interpolation formula using the points $[a, f(a)]$*,* $[b, f(b)]$*, and* $[c, f(c)]$ *is given by:*

$$\begin{aligned} x = &\frac{[y - f(a)][y - f(b)]c}{[f(c) - f(a)][f(c) - f(b)]} + \frac{[y - f(b)][y - f(c)]a}{[f(a) - f(b)][f(a) - f(c)]} \\ &+ \frac{[y - f(c)][y - f(a)]b}{[f(b) - f(c)][f(b) - f(a)]} \end{aligned} \tag{52}$$

Setting $y = 0$ *gives the next root estimate, which can be written more compactly as:*

$$x = b + \frac{P}{Q} \tag{53}$$

where, in terms of the intermediate variables $R = f(b)/f(c)$*,* $S = f(b)/f(a)$*, and* $T = f(a)/f(c)$*, the terms* P *and* Q *are defined as:*

$$P = S[T(R - T)(c - b) - (1 - R)(b - a)] \tag{54}$$

$$Q = (T - 1)(R - 1)(S - 1) \tag{55}$$

A key feature of Brent's method is its robustness. The algorithm includes contingency checks at each step. If the new estimate generated by Inverse Quadratic Interpolation is not satisfactory (e.g., it falls outside the bracketing interval $[a, b]$), the algorithm will reject it and fall back to a slower but guaranteed-to-converge bisection step. This hybrid approach ensures both speed and reliability, making it a highly effective and standard choice for numerical root-finding.

To apply Brent's method to the experiment, a target function, $g(\lambda)$, is defined such that its root corresponds to the Equivalent Hazard Rate, λ^*. This function is the difference between the CVA calculated with a trial hazard rate and the baseline CVA^*:

$$g(\lambda^*) = \text{CVA}_{\text{No IM}}(\lambda^*, \rho) - \text{CVA}^* = 0 \tag{56}$$

Brent's method is then employed to find the root of $g(\lambda^*)$, which is the specific hazard rate that satisfies the following constraint:

$$\text{CVA}_{\text{No IM}}(\lambda^*, \rho) = \text{CVA}_{\text{No IM}}(\lambda, \rho = 0) \equiv \text{CVA}^* \tag{57}$$

The difference, $\lambda - \lambda^*$, can subsequently be interpreted as the portion of the market's credit spread that was implicitly pricing in the risk now explicitly captured by ρ.

3. **Calculate and Compare Collateralised CVA**
Finally, we use this more nuanced risk allocation to price the CVA for portfolios that are collateralised with Initial Margin. For each value of ρ, two CVA calculations are performed using the method illustrated in Chapter 3:

   - The **"Original CVA"** (CVA_λ) is calculated using the high, fixed market hazard rate λ in conjunction with the explicit WWR correlation ρ.
   - The **"New CVA"** (CVA_{λ^*}) is calculated using the same WWR correlation ρ but with its corresponding Equivalent Hazard Rate, λ^*.

   By analysing the proportion $\text{CVA}_{\lambda^*}/\text{CVA}_\lambda$ under different levels of IM, we can investigate, after recalibrating the hazard rate, whether IM is able to cover the intended proportion of the losses through collateral. We can also quantify the extent to which a simple model with WWR might overstate the CVA for collateralised trades by failing to account for the WWR premium already embedded in the market's pricing of default risk. The results of this analysis for the GBM, Heston, and Merton models will be presented in the subsequent sections.

5.2 Analysis under the Geometric Brownian Motion

For the analysis presented in this and subsequent sections, a consistent set of parameters was used to define the simulated derivative contracts.

Contract and Market Parameters:

- **Maturity (T):** A maturity of one year was selected. While typical CDS maturities are longer (e.g., five years), a one-year horizon is a standard assumption in the context of regulatory capital calculations. This reflects the period over which a firm is assumed to be exposed before it can take action to mitigate or close out the risk.
- **Underlying Asset Spot Price (S_0):** Normalised to 100.
- **Moneyness:** Scenarios were defined by the strike price (K) relative to the spot price:
  - **At-the-Money (ATM):** $K = 100$
  - **In-the-Money (ITM):** $K = 80$ for call options and $K = 120$ for put options.
  - **Out-of-the-Money (OTM):** $K = 120$ for call options and $K = 80$ for put options.
- **Risk-Free Interest Rate (r):** 5%
- **Dividend Yield (q):** 0%
- **Market Hazard Rate (λ):** Market-observed hazard rate was set to 5%. According to Table 1 in [9], this spread is characteristic of a speculative-grade counterparty with a credit rating of Ba. This choice is deliberately made to reflect the high-risk profile of the leveraged entities, such as Archegos.
- **Recovery Rate (R):** 40%

While the absolute values of CVA is sensitive to the choice of λ and T, the core conclusion of this paper, which are derived from CVA ratios, is expected to remain robust within reasonable ranges of these parameters. When λT is small, the credit value adjustment roughly scales proportionally with λ as the default probability density ($\lambda e^{-\lambda T}$) is close to λ, and $e^{-\lambda T}$ is close to 1. Therefore, while variations in parameters does indeed affect the absolute magnitude of CVA figures, they do not cause significant changes in the relative amplification effect.

For this section, the volatility σ is assumed to be 0.2.

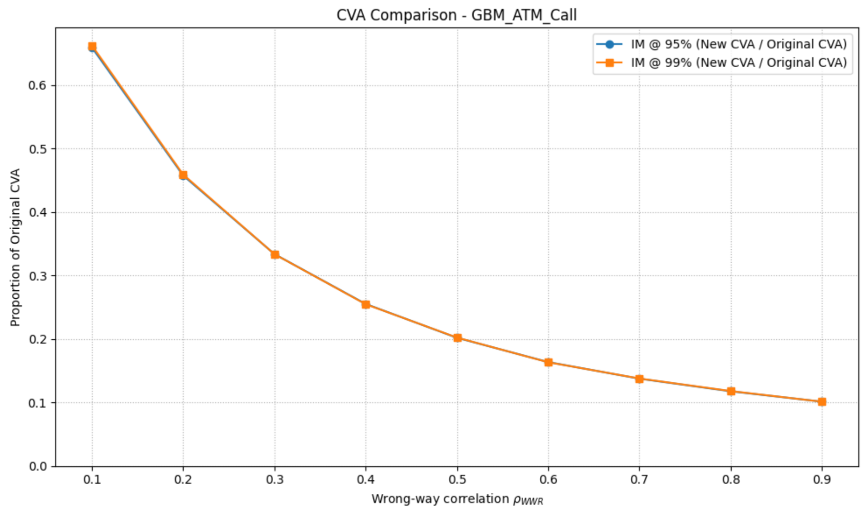

Figure 3: CVA Comparison for an ATM Call under the GBM model. The figure plots the ratio of CVA_{λ^*} to CVA_{λ} against the WWR correlation ρ

We begin the empirical analysis by examining the results under the simplest and most foundational framework: the Geometric Brownian Motion (GBM) model. As the benchmark process underpinning the Black-Scholes-Merton pricing paradigm, GBM provides a clear and tractable environment in which to test our central hypothesis. The purpose of this initial analysis is twofold: first, to establish a baseline result in a world of constant volatility and continuous price paths, and second, to test the robustness of our findings across different portfolio structures.

The experiment outlined in Section 5.1 was conducted for multiple portfolios, including call options, put options, and forward contracts at different levels of moneyness. The primary result of this analysis is the proportion $\text{CVA}_{\lambda^*}/\text{CVA}_{\lambda}$ as a function of the wrong-way risk correlation, ρ. For brevity, we present the At-the-Money Call option as the benchmark case in the main text; the results for all other scenarios, which were found to be qualitatively identical, are available in Appendix B.

The most immediate and striking result revealed by the analysis under the GBM model is the remarkable consistency observed across all tested scenarios. As illustrated in the figure, every portfolio structure – including out-of-the-money, at-the-money, and in-the-money call and put options, as well as at-the-money forward contracts – generates the same distinctive downward-sloping convex curve. This consistency provides robust and compelling evidence for the core hypothesis of this paper. Namely, it demonstrates that the central finding is not the product of a specific combination of derivatives and parameters, but rather stems from an intrinsic property of the experimental design itself – a ratio driven by the inverse relationship between the explicit WWR correlation coefficient ρ and the calibration equivalent risk rate λ^*. The fact that this result holds irrespective of the derivative's moneyness, direction, or payoff structure (non-linear vs. linear) strongly suggests that the potential overestimation of CVA by naive models is a general, structural issue within the GBM framework, not one confined to specific types of trades.

To quantify this effect, we first examine the theoretical amplification of CVA under a naive application of the WWR model. Table 3 shows that for a 99% IM portfolio, the CVA is amplified by a factor of 22.7 when the WWR correlation, ρ is 0.5. We note that for most real world scenarios, a realistic value of ρ will be between 0.25 to 0.5, so it is instructive to focus on these values instead of the extreme cases where $\rho = 1$. This dramatic figure substantiates the claim that conventional models, which ignore WWR, can severely underestimate CVA, as we have discussed in Section 5.1.

		CVA($\rho > 0$)/CVA($\rho = 0$)				
IM Amount	**CVA($\rho = 0$)**	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1.0}$
No IM	0.0283	1.5	2.6	5.0	7.9	11.2
IM @ 95%	0.0033	2.0	4.8	14.8	34.5	62.5
IM @ 99%	0.0011	2.2	6.1	22.7	63.7	139.4

Table 3: The multiplicative impact of leveraged WWR on CVA (GBM Model, ATM Call). The table shows the baseline CVA at $\rho = 0$ and the ratio CVA($\rho > 0$)/CVA($\rho = 0$) for various levels of WWR correlation, calculated using the fixed market hazard rate ($\lambda = 0.05$).

However, this dramatic CVA figure is based on the market hazard rate λ and embodies the potential "double counting" of risk, which was also mentioned in Section 5.1. The central finding of this research is revealed when we deconstruct the market's pricing of risk. Table 4 presents a more nuanced view, showing the CVA amplification calculated using the Equivalent Hazard Rate λ^*. The contrast is stark. After accounting for the implicit WWR premium in the market's credit spread, the residual amplification factor for the same 99% IM portfolio at $\rho = 0.5$ is only 4.6, compared to 22.7 when we used the market λ. We also note that the amplification factor under IM at 99% is larger than the factor under IM at 95% across all derivative types (see Appendix B for detailed illustration).

This comparison leads to a more balanced conclusion than that suggested by a naive application of the WWR model. Under the GBM model, while CVA is indeed underestimated by conventional models which do not include the effect of wrong way risk, the true underestimation for collateralised trades is likely not as severe as the headline amplification factors would suggest. The market's high price for credit, as reflected in elevated CDS spreads,

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$)				
		$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0283	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0033	1.3	1.9	3.0	4.4	5.6
IM @ 99%	0.0011	1.5	2.4	4.6	8.0	12.5

Table 4: The recalibrated multiplicative impact of leveraged WWR on CVA (GBM Model, ATM Call). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ. It represents the residual WWR effect after accounting for the implicit risk premium in the market hazard rate.

already contains a significant buffer for WWR. Our analysis, which separates the implicit market premium from the explicit model effect, suggests that the true, collateralised CVA, while higher than a simple model would predict, is not so catastrophically high as to preclude rational economic activity. This is further reinforced by the CVA comparison graphs, which consistently show the CVA_{λ^*} is only a fraction (approximately 10-20% at large ρ) of the CVA_{λ}, indicating that a naive model may be significantly overstating the true risk for these trades.

Another interesting pattern that the amplification factors show is that the magnitude of factors follows the descending order of: in-the-money (ITM), at-the-money (ATM) and out-of-the-money (OTM). In this paragraph, we will mainly showcase the effects of WWR after recalibration under $\rho = 0.5$, chosen due to its plausibility in the real world, but it is important to note that the trend mentioned below is also true for other values of ρ. From Table 5 and 6, we observe the exact trend mentioned, irrespective of the European derivative taken (Call or Put) or the IM amount. However, if we focus on the amplification factor when applied to a forward, its value is the highest among all cases. This suggests that the CVA under a forward contract is affected the most by wrong way risk under a fixed ρ. If our portfolio risk is predominantly governed by options risk, then the underestimation due to not treating wrong way risk in practice might not be as worse as one would imagine.

IM Amount	ATM	OTM	ITM
IM @ 95%	2.99	2.84	3.82
IM @ 99%	4.61	4.1	6.98

Table 5: The amplification factors calculated using Equivalent Hazard Rate λ^* under the GBM model, using the European Call option with $\rho = 0.5$

IM Amount	ATM	OTM	ITM
IM @ 95%	2.88	2.75	4.03
IM @ 99%	4.39	3.87	7.36

Table 6: The amplification factors calculated using Equivalent Hazard Rate λ^* under the GBM model, using the European Put option with $\rho = 0.5$

5.3 Analysis under the Heston model

In this section, we will focus on analysing the Heston stochastic volatility model, providing a more rigorous test of the dissertation's hypothesis under alternate market dynamics. For this section, the parameters used are:

- **Scaling Parameter** (σ): 1
- **Mean Reversion Speed** (κ): 1
- **Volatility of Variance** (η): 0.3
- **Correlation of** $dW_t^{(1)} dW_t^{(2)}$ (ρ_{Heston}): 0.7 or -0.7

We produced results for different scenarios, changing the parameters such as moneyness (in-the-money, out-of-the-money, at-the-money), derivatives (Call, Put, Forward) and the correlation between the Brownian Motion $\rho_{Heston} = dW_t^{(1)} dW_t^{(2)}$. For detailed results, please see Appendix C. Similar to the previous section, the graphs and tables are qualitatively similar to each other, so we will present the at-the-money call option with $\rho_{Heston} = 0.7$ as a baseline case. In general, the results are quite similar to that of the GBM model, but with slightly smaller values. In terms of the notations, since the Heston SDE uses the same notation ρ which defines the correlation between the two Brownian Motions, we will explicitly denote that as ρ_{Heston} below. ρ mentioned below is assumed to be the WWR correlation parameter.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0465	1.5	2.6	5.2	8.5	12.4
IM @ 95%	0.0069	2.0	4.7	14.2	32.6	58.7
IM @ 99%	0.0028	2.2	5.8	20.9	56.0	117.1

Table 7: The multiplicative impact of leveraged WWR on CVA (Heston Model, ATM Call, $\rho_{Heston} = 0.7$). The table shows the baseline CVA at $\rho = 0$ and the ratio CVA($\rho > 0$)/CVA($\rho = 0$) for various levels of WWR correlation, calculated using the fixed market hazard rate ($\lambda = 0.05$).

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0465	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0069	1.3	1.8	2.7	3.8	4.7
IM @ 99%	0.0028	1.4	2.2	4.0	6.6	9.5

Table 8: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ATM Call, $\rho_{Heston} = 0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

First, we examine the theoretical power of the WWR model when applied to the Heston framework. Table 7 quantifies this effect using a fixed market risk rate ($\lambda = 0.05$). Consequently, the CVA amplification effect introduced by stochastic volatility demonstrated a scale comparable to that observed in the GBM model. For the 99% IM portfolios, CVA is amplified by a factor of 20.9 at a WWR correlation of $\rho = 0.5$. Although the numerical results differ from the GBM model due to the different properties in the Heston model such as tail characteristics, this phenomenon again demonstrates that the dramatic theoretical impact of WWR is not an illusion of a simple constant volatility model, but rather persists in more realistic market dynamics such as the inclusion of stochastic volatility, as shown in the Heston model.

The purpose of this dissertation's experimental design is revealed, however, when we deconstruct this headline figure. The analysis in Table 8 calculates the amplification using the Equivalent Hazard Rate, λ^*, providing a more refined perspective. Similar to Section 5.2, this procedure allows us to isolate the residual WWR effect from the risk premium already embedded in the market's hazard rate. This procedure allows us to isolate the residual WWR effect from the risk premium already embedded in the market's hazard rate. The result is a dramatic reduction in the measured impact: the residual amplification factor for the same 99% IM portfolio at $\rho = 0.5$ is only 4. This finding reinforces the dissertation's central hypothesis: that the market's high price for credit, as reflected in elevated CDS spreads,functions as an efficient, albeit blunt, mechanism for pricing in complex dynamics such as stochastic volatility. The true CVA in collateralised transactions, while exceeding simplified model predictions, remains far from the catastrophic levels implied by a straightforward application of the WWR framework.

A unique observation for the Heston model is that the pattern of the amplification factors (after recalibration of λ^*) is dependent on the sign of ρ_{Heston}. Under a positive ρ_{Heston}, we observe that the amplification factor under European Call option is smaller than that of the European Put option under $\rho = 0.5$, given that moneyness is fixed. In contrast, when we take ρ_{Heston} to be negative, the opposite effect is observed; the amplification factor under the European Call Option is larger than that of the European call option, under the same parameters. Again, we note that although we only showcased results under $\rho = 0.5$, the results generated using other values of ρ are also consistent with the findings above. We should also note that the multiplier is largest for ITM options, followed by ATM, OTM options and forward contract. This is not dependent on the type of option, as both call and put option shows the same trend.

	ATM Call	**ATM Put**	**ITM Call**	**ITM Put**	**OTM Call**	**OTM Put**
IM @ 95%	2.7	2.8	3.5	4.2	2.6	2.6
IM @ 99%	4.0	4.3	6.0	8.6	3.4	3.5

Table 9: Amplification factors calculated using the Equivalent Hazard Rate λ^* under the Heston model, with $\rho = 0.5$ and $\rho_{Heston} = 0.7$.

	ATM Call	**ATM Put**	**ITM Call**	**ITM Put**	**OTM Call**	**OTM Put**
IM @ 95%	2.8	2.7	3.9	3.7	2.7	2.5
IM @ 99%	4.3	3.9	7.5	6.8	3.8	3.3

Table 10: Amplification factors calculated using the Equivalent Hazard Rate λ^* under the Heston model, with $\rho = 0.5$ and $\rho_{Heston} = -0.7$.

5.4 Analysis under the Merton model

In this section, we will start our analysis using the Merton Jump-Diffusion model (see Section 2.2.3), which allows us to test the dissertation's central hypothesis under a different setting which includes the possibility of discontinuous jump events. For this section, the parameters used are:

- **Jump Intensity (λ_{jump})**: 0.05
- **Diffusion Volatility (σ)**: 0.2
- **Volatility of Jump Size (δ)**: 0.25
- **Mean Jump Size (γ)**: 0.25 or -0.25

It is important to note the difference between λ_{jump}, which is the jump intensity in the Merton SDE, and the general λ, which is the hazard rate.

As before, the experiment was run across a comprehensive set of scenarios, including different portfolio types (Calls, Puts, Forwards), moneyness (ITM, ATM, OTM), and, most critically, different jump characteristics. In particular, we simulated the scenarios under both positive and negative values for the parameter γ, which models the direction of the jump (modelling both positive market shocks and negative market crashes).

Under the Merton jump diffusion model, the results are more flavourful and provide more insight into how the CVA behaves under different parameters. Table 11 and 12 shows the ratio of the CVA_{λ^*} in the presence of IM to CVA_{λ^*} without IM under an at-the-money call, with $\gamma = 0.25$ and -0.25. We want to compare the results of our experiment, as illustrated in Section 5.1 under a different light in order to investigate the effect of the smile on CVA_{λ^*}. We can see that, in this particular setup, the ratio is significantly larger for the case with positive γ, when compared to the case with negative γ. This is not an outlier case. In all other scenarios that we considered, including different moneyness and derivatives, we can observe the same trend (see Appendix D).

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	10%	16%	22.8%	52%	75%	91%
IM @ 99%	6.24%	10.83%	20.68%	43.42%	67.11%	86.96%

Table 11: Ratio of CVA_{λ^*} with Initial Margin to CVA_{λ^*} without Initial Margin (under the Merton Model, ATM Call, $\gamma = 0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

IM amount	CVA(with IM)/CVA(without IM) $\rho = 0$	$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1$
IM @ 95%	11%	15%	22%	37%	54%	69%
IM @ 99%	4.05%	6.08%	10.22%	21.04%	37.67%	56.13%

Table 12: Ratio of CVA_{λ^*} with Initial Margin to CVA_{λ^*} without Initial Margin (under the Merton Model, ATM Call, $\gamma = -0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

In Section 5.3, we mentioned that the amplification factor under the Heston model exhibits a trend which is dependent on the sign of ρ_{Heston} – that is, if there is an upward or downward sloping skew. Similarly, in the Merton model, we observed that the amplification factor is interlinked with the sign of γ, and hence direction of the jump. In Table 13 which represents a positive jump with $\gamma = 0.25$, for both ITM and OTM options, call options exhibit a higher CVA amplification than put options at low levels of ρ. However, this relationship inverts as the WWR correlation becomes strong, with put options then showing greater amplification. This entire pattern reverses in a negative jump scenario ($\gamma = -0.25$) as observed from Table 14. Here, call options show less amplification than puts at low ρ values, but greater amplification at high ρ values. This demonstrates that the specific payoff structure of an option dictates how its CVA is magnified by WWR, depending on whether the jump risk aligns with or opposes the tail exposure driving the correlated defaults. It is important to note that we only showed this trend for the case where Initial Margin is at 99%, but the same trend holds when Initial Margin is at 95%. Another interesting pattern observed in both Table 13 and 14 is that under both call and put options, the multiplier for ATM are larger that of ITM, followed by OTM and forward contract. This is consistent with our findings in both the GBM and Merton model.

Derivative & Moneyness	**CVA(with IM)/CVA(without IM)**				
	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
ATM Call	1.7	3.3	6.9	10.7	14.1
ATM Put	1.5	2.4	4.8	8.3	12.6
ITM Call	1.8	3.9	9.1	16.1	21.4
ITM Put	1.7	3.5	9.2	21.1	40.2
OTM Call	1.6	2.8	4.4	5.2	5.7
OTM Put	1.5	2.6	4.8	7.6	10.6
Forward	1.8	3.8	9.3	16.1	22.1

Table 13: Amplification Factors for different derivatives and moneyness calculated using the Equivalent Hazard rate λ^*, under the Merton Model with $\gamma = 0.25$ and Initial Margin @ 99%

Derivative & Moneyness	**CVA(with IM)/CVA(without IM)**				
	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
ATM Call	1.5	2.5	5.2	9.2	13.7
ATM Put	1.7	3.1	6.4	10.2	14.1
ITM Call	1.7	3.4	8.6	18.4	32.9
ITM Put	1.8	3.7	9.4	17.6	28.8
OTM Call	1.5	2.6	5.2	8.8	11.5
OTM Put	1.6	2.4	3.6	4.1	4.6
Forward	1.8	3.7	9.5	20.5	37.4

Table 14: Amplification Factors for different derivatives and moneyness calculated using the Equivalent Hazard rate λ^*, under the Merton Model with $\gamma = -0.25$ and Initial Margin @ 99%

6 Discussion and Conclusion

The motivation for this paper stems from the 2021 collapse of Archegos Capital Management. This crisis exposed major flaws in the derivatives regulatory framework established since 2008. It demonstrated that conventional credit valuation adjustment (CVA) models, which typically assume market risk and credit risk are mutually independent, cannot effectively handle leveraged WWR from counterparties. The core question of this research is whether the CVA is as severely underestimated as the results exhibited by contemporary WWR models, or whether the market has already priced in this tail risk through other mechanism, such as the market hazard rate.

To tackle this, we designed an experiment which assumes that the uncollaterised CVA ($\rho = 0$) can be accurately calculated. Our hypothesis is that while the CVA is almost certainly underpriced currently without taking WWR into account, it might not be as severe as practitioners, such as Pykhtin, would suggest. The methodology was structured to deconstruct the market's pricing of credit. First, we established a baseline CVA for an uncollateralised portfolio, assuming it represented the total risk budget priced by the market. Second, using this constraint, we recalibrated the market's "blunt" hazard rate to obtain a lower, recalibrated hazard rate λ^*. Finally, we tested our hypothesis by applying λ^* to portfolios under 95% and 99% IM, and compared our results with the CVA using the market hazard rate. This experiment was conducted across three processes – GBM, Heston, and Merton to ensure the robustness of the findings.

Key Findings

Across all underlying models, deconstructing this premium by using the recalibrated hazard rate λ^* resulted in a dramatic reduction of the CVA amplification factors. For a 99% IM portfolio under the GBM model, the residual amplification factor at $\rho = 0.5$ was only 4.6, a stark contrast to the naive amplification factor of 22.7. This confirms that simply layering an explicit WWR model on top of market-observed credit spreads leads to a significant overestimation of risk due to "double counting".

However, decomposing this premium does not eliminate the risk. The key conclusion of this study is that even if we assume that the CVA under no initial margin is priced appropriately, the CVA for collaterised portfolios (IM @ 95% and IM @ 99%) with moderate WWR effect ($0.25 \leq \rho \leq 0.5$) is several times higher than the CVA for portfolios without WWR. This conclusion holds regardless of the underlying process, including GBM, Heston and Merton model. Although differences in distribution, such as the skewness of the Heston model or the direction of jumps in the Merton model can affect the exact multiplier depending on the option payoff characteristics, it was not observed that the Heston or Merton models systematically produced significantly lower multipliers than the simplified GBM model.

This leads to two crucial insights. Firstly, while extreme CVA multipliers estimated by simply applying the Pykhtin model tend to be overestimated, CVA is still underestimated in collateralised transactions, and its impact is significant. Secondly, this underestimation of the CVA is not resolved merely by adopting more sophisticated fat-tailed models for the underlying asset distribution. The structural correlation of global risks (WWR) is, in fact, a unique and dominant risk factor.

Further Development and Limitations

While this dissertation provides a framework for deconstructing the WWR premium, we focused mostly on 3 derivatives (European Call, European Put, Forward) and 3 models (GBM, Heston, Merton) to ensure a clear and tractable analysis. This focus naturally presents several avenues for future research.

The first area for development lies in the choice of the dependency structure. This analysis employed the Gaussian copula for its simplicity as it only requires a single correlation parameter to link default events with market movements. However, the Gaussian copula has its limitations, especially when we are focused on capturing tail dependence, which is a critical feature of financial crises where correlations often intensify during extreme market stress. Future research could extend this framework by implementing alternative copulas, such as the Student's t-copula, which allows for explicit tail dependence. Another alternative is the Clayton copula, which is designed to capture stronger dependence in the lower tail, making it particularly useful for modelling joint-crash scenarios where two or more risk factors experience extreme negative events simultaneously. Such an extension would provide valuable insights into whether the CVA amplification effects are more pronounced when the joint-tail-risk dynamics of WWR are modelled more explicitly.

Another potential area for development would be to apply this experimental framework to a broader range of financial instruments. The dissertation deliberately focused on single-name equity derivatives to model the concentrated risk profile shown by the Archegos case. Consequently, instruments such as interest rate swaps and credit derivatives were excluded from our analysis. Applying the methodology outlined in Section 5.1 to an interest rate swap portfolio would test the WWR hypothesis in a market driven by macroeconomic factors. Similarly, analysing credit instruments could potentially reveal further effects on WWR. These extensions would allow for a more comprehensive understanding of how the implicit

WWR premium is included across different asset classes, and to tell us if the extent of CVA underestimation.

Bibliography

[1] Leif B. G. Andersen, Michael Pykhtin, and Alexander Sokol. Credit exposure in the presence of initial margin. *SSRN Electronic Journal*, 2016.

[2] Bank for International Settlements. Bis quarterly review, september 2021: Box c - the archegos debacle: a case of wrong-way risk? *BIS Quarterly Review*, pages 14–15, September 2021.

[3] BCBS and IOSCO. Margin requirements for non-centrally cleared derivatives. Technical report, Bank for International Settlements, 2015.

[4] Abdel Berkaoui, Mireille Bossy, and Awa Diop. Euler scheme for sdes with non-lipschitz diffusion coefficient: strong convergence. *ESAIM: Probability and Statistics*, 12:1–11, 2008.

[5] Richard P. Brent. *Algorithms for Minimization without Derivatives.* Prentice-Hall, Englewood Cliffs, NJ, 1973. Reprinted by Dover Publications, 2002.

[6] Rama Cont and Peter Tankov. *Financial Modelling with Jump Processes.* Chapman & Hall/CRC, Boca Raton, 2004.

[7] John C. Cox, Jonathan E. Ingersoll, and Stephen A. Ross. A theory of the term structure of interest rates. *Econometrica*, 53(2):385–407, 1985.

[8] A. Dickinson. Mind the gap. *Risk*, 2022.

[9] John Hull, Mirela Predescu, and Alan White. Bond prices, default probabilities and risk premiums. *SSRN Electronic Journal*, 2013.

[10] Peter E. Kloeden and Eckhard Platen. *Numerical Solution of Stochastic Differential Equations*, volume 23 of *Applications of Mathematics.* Springer, Berlin, Heidelberg, 1992.

[11] Roger Lord, Remmert Koekkoek, and Dick Van Dijk. A comparison of biased simulation schemes for stochastic volatility models. *Quantitative Finance*, 10(2):177–194, 2010.

[12] Kazuhisa Matsuda. Introduction to merton jump diffusion model. Technical report, Max Matsuda Research, n.d. Accessed: 2025-09-15.

[13] William H. Press, Saul A. Teukolsky, William T. Vetterling, and Brian P. Flannery. *Numerical Recipes in FORTRAN: The Art of Scientific Computing*. Cambridge University Press, Cambridge, 2 edition, 1992.

[14] Michael Pykhtin. Weighting for leverage. *Risk*, 2024.

A Expected Exposure Results for other Underlying Models

This appendix provides supplementary results concerning geometric Brownian motion (GBM) for the expected exposure (EE) analysis discussed in Chapter 4. The parameter settings employed are identical to those described in the main text.

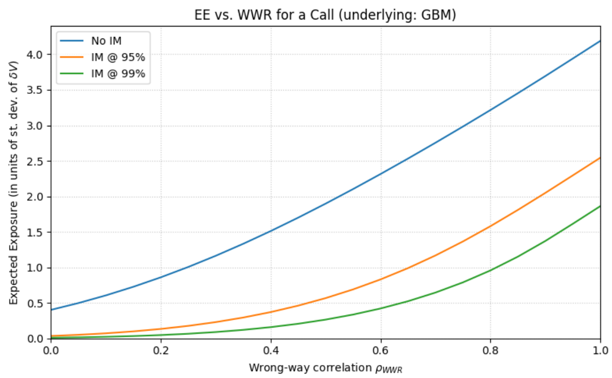

Figure 4: Expected Exposure against ρ for the GBM ATM Call model with $h(t) = 0.001$ and strike $K = 100$.

B Tables for the GBM model

IM Amount	**CVA($\rho = 0$)**	**CVA($\rho > 0$)/CVA($\rho = 0$)**				
		$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0198	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0024	1.30	1.83	2.88	4.13	5.30
IM @ 99%	0.0009	1.44	2.33	4.39	7.50	11.60

Table 15: The recalibrated multiplicative impact of leveraged WWR on CVA (GBM Model, ATM Put). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ. It represents the residual WWR effect after accounting for the implicit risk premium in the market hazard rate.

IM Amount	**CVA($\rho = 0$)**	**CVA($\rho > 0$)/CVA($\rho = 0$)**				
		$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0455	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0028	1.41	2.18	3.96	6.33	8.90
IM @ 99%	0.0005	1.64	3.11	7.43	15.13	29.47

Table 16: The recalibrated multiplicative impact of leveraged WWR on CVA (GBM Model, Forward). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ. It represents the residual WWR effect after accounting for the implicit risk premium in the market hazard rate.

IM Amount	**CVA($\rho = 0$)**	**CVA($\rho > 0$)/CVA($\rho = 0$)**				
		$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0447	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0029	1.39	2.14	3.82	6.15	8.47
IM @ 99%	0.0006	1.62	3.03	6.98	14.55	27.19

Table 17: The recalibrated multiplicative impact of leveraged WWR on CVA (GBM Model, ITM Call). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ. It represents the residual WWR effect after accounting for the implicit risk premium in the market hazard rate.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0397	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0022	1.41	2.19	4.03	6.56	9.35
IM @ 99%	0.0004	1.65	3.18	7.82	16.53	33.85

Table 18: The recalibrated multiplicative impact of leveraged WWR on CVA (GBM Model, ITM Put). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ. It represents the residual WWR effect after accounting for the implicit risk premium in the market hazard rate.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0063	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0009	1.31	1.83	2.84	4.01	5.02
IM @ 99%	0.0004	1.44	2.25	4.10	6.69	9.64

Table 19: The recalibrated multiplicative impact of leveraged WWR on CVA (GBM Model, OTM Call). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ. It represents the residual WWR effect after accounting for the implicit risk premium in the market hazard rate.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0015	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0002	1.30	1.79	2.75	3.79	4.67
IM @ 99%	0.0001	1.42	2.18	3.87	6.07	8.47

Table 20: The recalibrated multiplicative impact of leveraged WWR on CVA (GBM Model, OTM Put). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ. It represents the residual WWR effect after accounting for the implicit risk premium in the market hazard rate.

C Tables for the Heston model

IM Amount	$\mathbf{CVA}(\rho = 0)$	$\mathbf{CVA}(\rho > 0)/\mathbf{CVA}(\rho = 0)$				
		$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0335	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0044	1.3	1.8	2.8	3.9	4.8
IM @ 99%	0.0016	1.4	2.3	4.3	7.2	10.6

Table 21: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ATM Put, $\rho_{Heston} = 0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	$\mathbf{CVA}(\rho = 0)$	$\mathbf{CVA}(\rho > 0)/\mathbf{CVA}(\rho = 0)$				
		$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0671	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0055	1.4	2.0	3.5	5.4	7.3
IM @ 99%	0.0015	1.6	2.8	6.0	11.6	19.8

Table 22: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ITM Call, $\rho_{Heston} = 0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	$\mathbf{CVA}(\rho = 0)$	$\mathbf{CVA}(\rho > 0)/\mathbf{CVA}(\rho = 0)$				
		$\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0606	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0029	1.4	2.3	4.2	7.0	10.0
IM @ 99%	0.0004	1.7	3.4	8.6	19.3	40.8

Table 23: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ITM Put, $\rho_{Heston} = 0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	**CVA($\rho = 0$)**	**CVA($\rho > 0$)/CVA($\rho = 0$)** $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0119	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0024	1.3	1.7	2.6	3.4	4.1
IM @ 99%	0.0013	1.4	2.1	3.4	5.1	6.7

Table 24: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, OTM Call, $\rho_{Heston} = 0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	**CVA($\rho = 0$)**	**CVA($\rho > 0$)/CVA($\rho = 0$)** $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0029	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0006	1.3	1.7	2.6	3.5	4.1
IM @ 99%	0.0003	1.4	2.1	3.5	5.3	6.9

Table 25: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, OTM Put, $\rho_{Heston} = 0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	**CVA($\rho = 0$)**	**CVA($\rho > 0$)/CVA($\rho = 0$)** $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0458	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0058	1.3	1.8	2.8	4.1	5.1
IM @ 99%	0.0021	1.4	2.3	4.3	7.5	11.1

Table 26: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ATM Call, $\rho_{Heston} = -0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	**CVA($\rho = 0$)**	**CVA($\rho > 0$)/CVA($\rho = 0$)** $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0341	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0052	1.3	1.8	2.7	3.7	4.5
IM @ 99%	0.0021	1.4	2.2	3.9	6.4	9.1

Table 27: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ATM Put, $\rho_{Heston} = -0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0670	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0041	1.4	2.2	3.9	6.3	8.8
IM @ 99%	0.0008	1.7	3.1	7.5	15.7	30.4

Table 28: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ITM Call, $\rho_{Heston} = -0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0607	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0041	1.4	2.1	3.7	5.9	8.1
IM @ 99%	0.0008	1.6	3.0	6.8	13.9	25.6

Table 29: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, ITM Put, $\rho_{Heston} = -0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0114	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0020	1.3	1.8	2.7	3.6	4.5
IM @ 99%	0.0009	1.4	2.2	3.8	5.8	8.0

Table 30: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, OTM Call, $\rho_{Heston} = -0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

IM Amount	CVA($\rho = 0$)	CVA($\rho > 0$)/CVA($\rho = 0$) $\rho = 0.1$	$\rho = 0.25$	$\rho = 0.5$	$\rho = 0.75$	$\rho = 1.0$
No IM	0.0031	N/A	N/A	N/A	N/A	N/A
IM @ 95%	0.0007	1.3	1.7	2.5	3.3	3.9
IM @ 99%	0.0004	1.4	2.0	3.3	4.8	6.1

Table 31: The recalibrated multiplicative impact of leveraged WWR on CVA (Heston Model, OTM Put, $\rho_{Heston} = -0.7$). This table shows the same amplification ratio, but is calculated using the Equivalent Hazard Rate, λ^*, for each value of ρ.

D Graphs and Tables for the Merton model

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	12%	16%	23%	37%	54%	68%
IM @ 99%	4.33%	6.4%	10.56%	20.97%	36.58%	54.94%

Table 32: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, ATM Put, $\gamma = 0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	12%	17%	28%	49%	72%	89%
IM @ 99%	6.03%	10.17%	18.56%	38.48%	61.99%	84.50%

Table 33: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, ATM Put, $\gamma = -0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	7%	12%	21%	44%	69%	88%
IM @ 99%	3.80%	6.96%	14.64%	34.79%	59.79%	82.41%

Table 34: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, ITM Call, $\gamma = 0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	6%	9%	14%	26%	44%	60%
IM @ 99%	1.35%	2.30%	4.63%	11.57%	24.79%	43.69%

Table 35: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, ITM Call, $\gamma = -0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	5%	8%	12%	23%	40%	56%
IM @ 99%	0.94%	1.62%	3.34%	8.70%	19.85%	37.59%

Table 36: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, ITM Put, $\gamma = 0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	7%	10%	18%	36%	60%	83%
IM @ 99%	2.65%	4.83%	9.95%	25.14%	47.27%	75.56%

Table 37: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, ITM Put, $\gamma = -0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	16%	27%	45%	69%	84%	93%
IM @ 99%	5.00%	24.79%	41.95%	66.01%	80.68%	90.35%

Table 38: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, OTM Call, $\gamma = 0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	14%	18%	27%	45%	65%	79%
IM @ 99%	6.10%	9.15%	15.93%	31.39%	52.92%	69.97%

Table 39: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, OTM Call, $\gamma = -0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	15%	20%	30%	48%	68%	81%
IM @ 99%	7.15%	10.83%	18.42%	35.07%	57.11%	73.81%

Table 40: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, OTM Put, $\gamma = 0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

	CVA(with IM)/CVA(without IM)					
IM amount	$\boldsymbol{\rho = 0}$	$\boldsymbol{\rho = 0.1}$	$\boldsymbol{\rho = 0.25}$	$\boldsymbol{\rho = 0.5}$	$\boldsymbol{\rho = 0.75}$	$\boldsymbol{\rho = 1}$
IM @ 95%	22%	34%	52%	75%	86%	94%
IM @ 99%	20.21%	32.37%	49.72%	71.85%	83.60%	91.07%

Table 41: Ratio of CVA with Initial Margin to CVA without Initial Margin (under the Merton Model, OTM Put, $\gamma = -0.25$), shown for various levels of correlation (ρ) and IM levels (95% and 99%) using the Equivalent Hazard Rate, λ^*

E Source Code

E.1 Functions

```
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt
from scipy.stats import norm, rankdata
from functools import partial
from scipy.optimize import brentq
import pandas as pd
import os



# Pricing Functions
def bs_call_price(S, K, T, r, q, sigma):
    if T <= 1e-6: return np.maximum(S - K, 0.0)
    d1 = (np.log(S / K) + (r - q + 0.5 * sigma ** 2) * T) / (sigma *
        np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return np.exp(-q * T) * S * norm.cdf(d1) - np.exp(-r * T) * K *
       norm.cdf(d2)


def bs_put_price(S, K, T, r, q, sigma):
    if T <= 1e-6: return np.maximum(K - S, 0.0)
    d1 = (np.log(S / K) + (r - q + 0.5 * sigma ** 2) * T) / (sigma *
        np.sqrt(T))
    d2 = d1 - sigma * np.sqrt(T)
    return np.exp(-r * T) * K * norm.cdf(-d2) - np.exp(-q * T) * S *
        norm.cdf(-d1)


def forward_price(S, K, T, r, q, **kwargs):
    return (S * np.exp(-q * T) - K * np.exp(-r * T))

```

```python

# Underlying Assets
def generate_gbm_paths(S0, r, q, sigma, n_paths, mpor_duration_years, rng, **kwargs):
    drift = (r - q - 0.5 * sigma ** 2) * mpor_duration_years
    diffusion = sigma * np.sqrt(mpor_duration_years)
    return S0 * np.exp(drift + diffusion * rng.standard_normal(n_paths))


def generate_merton_paths(S0, r, sigma, lam, gamma, delta_jump, n_paths, mpor_duration_years, rng, q=0.0, **kwargs):
    kbar = np.exp(gamma + 0.5 * delta_jump ** 2) - 1
    drift = (r - q - lam * kbar - 0.5 * sigma ** 2) * mpor_duration_years
    diffusion_coeff = sigma * np.sqrt(mpor_duration_years)
    Z = rng.standard_normal(n_paths)
    brownian_component = np.exp(drift + diffusion_coeff * Z)
    num_jumps = rng.poisson(lam * mpor_duration_years, n_paths)
    total_jump_factor = np.ones(n_paths)
    for i in range(n_paths):
        if num_jumps[i] > 0:
            jumps_log = rng.normal(gamma, delta_jump, num_jumps[i])
            total_jump_factor[i] = np.exp(np.sum(jumps_log))
    return S0 * brownian_component * total_jump_factor


def generate_heston_paths(S0, z0, r, sigma, kappa, eta, rho, n_paths, mpor_days, rng, q=0.0, **kwargs):
    """ Simulates final stock prices using Heston model. """
    dt = 1 / 252.0
    num_steps = mpor_days
    S = np.zeros((num_steps + 1, n_paths));
    S[0, :] = S0
    z = np.zeros((num_steps + 1, n_paths));
    z[0, :] = z0
    for i in range(num_steps):
```

```
        alpha, beta = rng.standard_normal(n_paths), rng.standard_normal(n_paths)
        z_max = np.maximum(z[i, :], 0)
        z[i + 1, :] = (z[i, :] + kappa * (1 - z[i, :]) * dt +
                       eta * (rho * alpha + np.sqrt(1 - rho ** 2) * beta) * np.sqrt(z_max) * np.sqrt(dt))
        S[i + 1, :] = S[i, :] * np.exp((r - q - 0.5 * sigma ** 2 * z_max) * dt +
                                       sigma * np.sqrt(z_max) * alpha * np.sqrt(dt))
    return S[-1, :]


# Portfolio Increment
def generate_derivative_increments(underlying_path_generator, pricing_function, **params):
    pricing_args_initial = {
        'S': params['S0'], 'K': params['K'], 'T': params['T_current'], 'r': params['r'],
        'q': params['q'], 'sigma': params['pricing_sigma']
    }
    V_initial = pricing_function(**pricing_args_initial)

    S_final = underlying_path_generator(**params)
    pricing_args_final = pricing_args_initial.copy()
    pricing_args_final['S'] = S_final
    pricing_args_final['T'] = params['T_current'] - params['mpor_duration_years']
    V_final = pricing_function(**pricing_args_final)
    return V_final - V_initial


def build_Z(delta_V):
    ranks = rankdata(delta_V, method="average");
    N = delta_V.size
    u = (ranks - 0.5) / N;
    eps = 0.5 / N
    return norm.ppf(np.clip(u, eps, 1.0 - eps))
```

```


def weights(z, rho_wwr, h):
    denom = np.sqrt(max(1.0 - rho_wwr ** 2, 1e-9))
    arg = (norm.ppf(h) + rho_wwr * z) / denom
    return norm.cdf(arg) / h


def expected_exposure(delta_V, rho_wwr, IM, h):
    Z = build_Z(delta_V)
    w = weights(Z, rho_wwr, h)
    exposure = np.maximum(delta_V - IM, 0.0)
    return np.mean(w * exposure)

# Calculate CVA
def calculate_cva(rho_wwr, im_levels, params, delta_v_generator):
    h, R, hazard, r = params['h'], params['R'], params['hazard'], params['r']
    time_grid = np.linspace(1 / 252, params['T_maturity'], 25)
    params['mpor_duration_years'] = params['mpor_days'] / 252.0

    ee_profiles = {lbl: [] for lbl in im_levels}
    for t in time_grid:
        params['T_current'] = params['T_maturity'] - t
        delta_V_t = delta_v_generator(**params)
        current_sigma_dv = np.std(delta_V_t)
        for label, z_score in im_levels.items():
            IM = 0.0 if z_score == 0.0 else z_score * current_sigma_dv
            EE_t = expected_exposure(delta_V_t, rho_wwr, IM, h)
            ee_profiles[label].append(EE_t)

    dPD = np.diff(np.insert(1 - np.exp(-hazard * time_grid), 0, 0.0))
    disc = np.exp(-r * time_grid)
    cva_results = {}
    for label, ee_series in ee_profiles.items():
```

```
        cva_results[label] = (1 - R) * np.dot(np.array(ee_series) *
            disc, dPD)
    return cva_results
```

E.2 Main Script: Create Expected Exposure vs ρ Graph

```
        if __name__ == "__main__":

    # 1. CHOOSE PORTFOLIO AND UNDERLYING MODEL
    PORTFOLIO_TYPE = 'Call'  # Options: 'Call', 'Put', 'Forward'
    UNDERLYING_MODEL = 'Merton'  # Options: 'GBM', 'Merton', 'Heston
       '

    # 2. DEFINE PARAMETERS
    common_params = {
        'S0': 100, 'K': 100, 'T_maturity': 1.0, 'r': 0.05, 'q': 0.0,
        'n_paths': 200000, 'mpor_days': 10, 'h': 0.001,
        'rng': np.random.default_rng(0)
    }
    gbm_params = {'sigma': 0.2}
    merton_params = {'sigma': 0.2, 'lam': 0.05, 'gamma': 0.25, '
       delta_jump': 0.25}
    heston_params = {'z0': 0.04, 'sigma': 1.0, 'kappa': 3.0, 'eta':
       0.3, 'rho': -0.7}

    # 3. FUNCTIONS AND MERGE PARAMETERS
    portfolio_map = {
        'Call': bs_call_price, 'Put': bs_put_price, 'Forward':
            forward_price
    }
    pricing_function = portfolio_map.get(PORTFOLIO_TYPE)

    model_map = {
        'GBM': (generate_gbm_paths, gbm_params),
        'Merton': (generate_merton_paths, merton_params),
        'Heston': (generate_heston_paths, heston_params)
    }
```

```
    underlying_generator, model_specific_params = model_map.get(UNDERLYING_MODEL)

    if not pricing_function or not underlying_generator:
        raise ValueError("Invalid PORTFOLIO_TYPE or UNDERLYING_MODEL selected.")

    params = {**common_params, **model_specific_params}


    if UNDERLYING_MODEL == 'Heston':
        params['pricing_sigma'] = np.sqrt(params['z0'])
    else:
        params['pricing_sigma'] = params['sigma']

    params['mpor_duration_years'] = params['mpor_days'] / 252.0
    # For a single point EE analysis, we can pick a representative time, e.g., t=0
    params['T_current'] = params['T_maturity']

    delta_v_generator = partial(generate_derivative_increments, underlying_generator, pricing_function)

    # 4. GENERATE THE DELTA_V DISTRIBUTION ONCE
    print(f"Generating delta_V for a {PORTFOLIO_TYPE} using the {UNDERLYING_MODEL} model...")
    delta_v_dist = delta_v_generator(**params)
    delta_v_std_dev = np.std(delta_v_dist)
    print(f"Standard deviation of delta_V is: {delta_v_std_dev:.4f}")

    # 5. CALCULATE EE FOR A RANGE OF RHO VALUES
    im_levels = {'No IM': 0.0, 'IM @ 95%': norm.ppf(0.95), 'IM @ 99%': norm.ppf(0.99)}
    rhos_wwr = np.linspace(0, 1.0, 21)
    ee_results_normalized = {lbl: [] for lbl in im_levels}

    print("\nCalculating Expected Exposure for each rho...")
```

```python
    for rho in rhos_wwr:
        for label, z_score in im_levels.items():
            IM = 0.0 if z_score == 0.0 else z_score * delta_v_std_dev
            ee = expected_exposure(delta_v_dist, rho, IM, params['h'])
            ee_normalized = ee / delta_v_std_dev
            ee_results_normalized[label].append(ee_normalized)
    print("Calculation complete.")

    # 6. PLOT THE RESULTS
    plt.figure(figsize=(8, 5))
    plt.plot(rhos_wwr, ee_results_normalized['No IM'], label='No IM')
    plt.plot(rhos_wwr, ee_results_normalized['IM @ 95%'], label='IM @ 95%')
    plt.plot(rhos_wwr, ee_results_normalized['IM @ 99%'], label='IM @ 99%')

    plt.xlabel(r'Wrong-way correlation $\rho_{WWR}$')
    plt.ylabel('Expected Exposure (in units of st. dev. of $\delta V$)')
    plt.title(f'EE vs. WWR for a {PORTFOLIO_TYPE} (underlying: {UNDERLYING_MODEL})')
    plt.legend()
    plt.grid(True, linestyle=':', alpha=0.7)
    plt.xlim(0, 1)
    plt.ylim(bottom=0)
    plt.tight_layout()
    plt.show()
```

E.3 Main Script: Calculating the Equivalent Hazard Rate λ^* and CVA_{λ^*}

```python
        if __name__ == "__main__":

    # 1. SETUP
    PORTFOLIO_TYPE = 'Call'
```

```python
    UNDERLYING_MODEL = 'Merton'
    BASE_HAZARD_RATE = 0.05

    # Parmeters
    common_params = {
        'S0': 100.0, 'K': 100.0, 'T_maturity': 1.0, 'r': 0.05, 'q': 0.0,
        'n_paths': 100000, 'mpor_days': 10, 'h': 0.001, 'R': 0.4,
        'rng': np.random.default_rng(0)
    }
    gbm_params = {'sigma': 0.2}
    merton_params = {'sigma': 0.2, 'lam': 0.05, 'gamma': 0.25, 'delta_jump': 0.25}
    heston_params = {'z0': 0.04, 'sigma': 1.0, 'kappa': 3.0, 'eta': 0.3, 'rho': -0.7}

    # Model and combine parameters
    portfolio_map = {'Call': bs_call_price, 'Put': bs_put_price, 'Forward': forward_price}
    model_map = {'GBM': (generate_gbm_paths, gbm_params), 'Merton': (generate_merton_paths, merton_params),
                 'Heston': (generate_heston_paths, heston_params)}

    pricing_function = portfolio_map.get(PORTFOLIO_TYPE)
    underlying_generator, model_specific_params = model_map.get(UNDERLYING_MODEL)
    params = {**common_params, **model_specific_params, 'hazard': BASE_HAZARD_RATE}

    if UNDERLYING_MODEL == 'Heston':
        params['pricing_sigma'] = np.sqrt(params['z0'])
    else:
        params['pricing_sigma'] = params['sigma']

    delta_v_generator = partial(generate_derivative_increments, underlying_generator, pricing_function)

    # 2. CALCULATE ORIGINAL CVA VALUES (FIXED LAMBDA)
```

```
    im_levels = {'No IM': 0.0, 'IM @ 95%': norm.ppf(0.95), 'IM @ 99%': norm.ppf(0.99)}
    rhos_wwr = np.linspace(0.1, 0.9, 9)

    print("Step 1: Calculating original CVA values with fixed lambda...")
    original_cva_results = {lbl: [] for lbl in im_levels}
    for rho in rhos_wwr:
        cva_out = calculate_cva(rho, im_levels, params, delta_v_generator)
        for lbl in im_levels:
            original_cva_results[lbl].append(cva_out[lbl])

    # 3. ESTABLISH BASELINE CVA*
    cva_star_dict = calculate_cva(0.0, im_levels, params, delta_v_generator)
    CVA_STAR = cva_star_dict['No IM']
    print(f"\nBaseline CVA* (No IM, rho=0, lambda={BASE_HAZARD_RATE}) = {CVA_STAR:.5f}")

    # 4. SOLVE FOR LAMBDA* AND CALCULATE NEW IM CVA VALUES
    print("\nStep 2: Solving for equivalent lambda* for each rho...")

    def cva_solver(lambda_trial, rho_val, target_cva):
        current_params = params.copy()
        current_params['hazard'] = lambda_trial
        cva_out = calculate_cva(rho_val, {'No IM': 0.0}, current_params, delta_v_generator)
        return cva_out['No IM'] - target_cva

    equivalent_lambdas = []
    new_cva_results = {lbl: [] for lbl in ['IM @ 95%', 'IM @ 99%']}

    for rho in rhos_wwr:
        try:
            lambda_star = brentq(cva_solver, 1e-6, 1.0, args=(rho, CVA_STAR))
```

```
            equivalent_lambdas.append(lambda_star)
            print(f"  For rho = {rho:.2f}, equivalent lambda* = {lambda_star:.5f}")

            params_new_lambda = params.copy()
            params_new_lambda['hazard'] = lambda_star
            cva_out_new = calculate_cva(rho, im_levels, params_new_lambda, delta_v_generator)
            for lbl in ['IM @ 95%', 'IM @ 99%']:
                new_cva_results[lbl].append(cva_out_new[lbl])

        except ValueError:
            print(f"  Could not find a solution for rho = {rho:.2f}.")
            equivalent_lambdas.append(np.nan)
            for lbl in ['IM @ 95%', 'IM @ 99%']:
                new_cva_results[lbl].append(np.nan)

    # 5. COMPILE AND DISPLAY RESULTS
    print("\n--- Final Comparison ---")
    results_df = pd.DataFrame({
        'rho_WWR': rhos_wwr,
        'Equivalent_Lambda_Star': equivalent_lambdas,
        'Original_CVA_95': original_cva_results['IM @ 95%'],
        'New_CVA_95': new_cva_results['IM @ 95%'],
        'Original_CVA_99': original_cva_results['IM @ 99%'],
        'New_CVA_99': new_cva_results['IM @ 99%']
    })

    results_df['Proportion_95'] = results_df['New_CVA_95'] / results_df['Original_CVA_95']
    results_df['Proportion_99'] = results_df['New_CVA_99'] / results_df['Original_CVA_99']

    pd.set_option('display.float_format', '{:.5f}'.format)
    print(results_df)

    # Plot
```

```python
    plt.figure(figsize=(10, 6))
    plt.plot(results_df['rho_WWR'], results_df['Proportion_95'], 'o-', label='IM @ 95% (New CVA / Original CVA)')
    plt.plot(results_df['rho_WWR'], results_df['Proportion_99'], 's-', label='IM @ 99% (New CVA / Original CVA)')
    plt.xlabel(r'Wrong-way correlation $\rho_{WWR}$')
    plt.ylabel('Proportion of Original CVA')
    plt.title('Comparison of CVA with Fixed vs. Equivalent Hazard Rate')
    plt.legend()
    plt.grid(True, linestyle=':')
    plt.ylim(bottom=0)
    plt.tight_layout()
    plt.show()
```

E.4 Main Script: Creating Tables with under different models and parameters

```python
        if __name__ == "__main__":

    # --- 1. DEFINE ALL SCENARIOS TO RUN ---
    scenarios = [
        # --- CALL OPTIONS ---
        # GBM Scenarios
        {'name': 'GBM_OTM_Call', 'model': 'GBM', 'portfolio': 'Call', 'K': 120},
        {'name': 'GBM_ATM_Call', 'model': 'GBM', 'portfolio': 'Call', 'K': 100},
        {'name': 'GBM_ITM_Call', 'model': 'GBM', 'portfolio': 'Call', 'K': 80},
        {'name': 'GBM_ITM_Put', 'model': 'GBM', 'portfolio': 'Put', 'K': 120},
        {'name': 'GBM_ATM_Put', 'model': 'GBM', 'portfolio': 'Put', 'K': 100},
        {'name': 'GBM_OTM_Put', 'model': 'GBM', 'portfolio': 'Put', 'K': 80},
        {'name': 'GBM_Forward', 'model': 'GBM', 'portfolio': 'Forward', 'K': 100},
```

```
        # Heston Scenarios (Negative and Positive Correlation)
        {'name': 'Heston_NegRho_OTM_Call', 'model': 'Heston', 'portfolio': 'Call', 'K': 120, 'rho': -0.7},
        {'name': 'Heston_NegRho_ATM_Call', 'model': 'Heston', 'portfolio': 'Call', 'K': 100, 'rho': -0.7},
        {'name': 'Heston_NegRho_ITM_Call', 'model': 'Heston', 'portfolio': 'Call', 'K': 80, 'rho': -0.7},
        {'name': 'Heston_PosRho_OTM_Call', 'model': 'Heston', 'portfolio': 'Call', 'K': 120, 'rho': 0.7},
        {'name': 'Heston_PosRho_ATM_Call', 'model': 'Heston', 'portfolio': 'Call', 'K': 100, 'rho': 0.7},
        {'name': 'Heston_PosRho_ITM_Call', 'model': 'Heston', 'portfolio': 'Call', 'K': 80, 'rho': 0.7},

        {'name': 'Heston_NegRho_ITM_Put', 'model': 'Heston', 'portfolio': 'Put', 'K': 120, 'rho': -0.7},
        {'name': 'Heston_NegRho_ATM_Put', 'model': 'Heston', 'portfolio': 'Put', 'K': 100, 'rho': -0.7},
        {'name': 'Heston_NegRho_OTM_Put', 'model': 'Heston', 'portfolio': 'Put', 'K': 80, 'rho': -0.7},
        {'name': 'Heston_PosRho_ITM_Put', 'model': 'Heston', 'portfolio': 'Put', 'K': 120, 'rho': 0.7},
        {'name': 'Heston_PosRho_ATM_Put', 'model': 'Heston', 'portfolio': 'Put', 'K': 100, 'rho': 0.7},
        {'name': 'Heston_PosRho_OTM_Put', 'model': 'Heston', 'portfolio': 'Put', 'K': 80, 'rho': 0.7},

        {'name': 'Heston_PosRho_Forward', 'model': 'Heston', 'portfolio': 'Forward', 'K': 100, 'rho': 0.7},
        {'name': 'Heston_NegRho_Forward', 'model': 'Heston', 'portfolio': 'Forward', 'K': 100, 'rho': -0.7},
        # Merton Scenarios (Positive and Negative Jumps)
        {'name': 'Merton_PosJump_ATM_Call', 'model': 'Merton', 'portfolio': 'Call', 'K': 100, 'gamma': 0.25},
        {'name': 'Merton_PosJump_OTM_Call', 'model': 'Merton', 'portfolio': 'Call', 'K': 120, 'gamma': 0.25},
        {'name': 'Merton_PosJump_ITM_Call', 'model': 'Merton', 'portfolio': 'Call', 'K': 80, 'gamma': 0.25},
```

```
        {'name': 'Merton_NegJump_OTM_Call', 'model': 'Merton', 'portfolio': 'Call', 'K': 120, 'gamma': -0.25},
        {'name': 'Merton_NegJump_ITM_Call', 'model': 'Merton', 'portfolio': 'Call', 'K': 80, 'gamma': -0.25},
        {'name': 'Merton_NegJump_ATM_Call', 'model': 'Merton', 'portfolio': 'Call', 'K': 100, 'gamma': -0.25},

        {'name': 'Merton_PosJump_ATM_Put', 'model': 'Merton', 'portfolio': 'Put', 'K': 100, 'gamma': 0.25},
        {'name': 'Merton_PosJump_OTM_Put', 'model': 'Merton', 'portfolio': 'Put', 'K': 80, 'gamma': 0.25},
        {'name': 'Merton_PosJump_ITM_Put', 'model': 'Merton', 'portfolio': 'Put', 'K': 120, 'gamma': 0.25},
        {'name': 'Merton_NegJump_OTM_Put', 'model': 'Merton', 'portfolio': 'Put', 'K': 80, 'gamma': -0.25},
        {'name': 'Merton_NegJump_ITM_Put', 'model': 'Merton', 'portfolio': 'Put', 'K': 120, 'gamma': -0.25},
        {'name': 'Merton_NegJump_ATM_Put', 'model': 'Merton', 'portfolio': 'Put', 'K': 100, 'gamma': -0.25},

        {'name': 'Merton_PosJump_forward', 'model': 'Merton', 'portfolio': 'Forward', 'K': 100, 'gamma': 0.25},
        {'name': 'Merton_NegJump_forward', 'model': 'Merton', 'portfolio': 'Forward', 'K': 100, 'gamma': -0.25},
    ]


    plot_output_dir = "cva_comparison_plots"
    table_output_dir = "cva_results_tables"
    if not os.path.exists(plot_output_dir): os.makedirs(plot_output_dir)
    if not os.path.exists(table_output_dir): os.makedirs(table_output_dir)

    for scenario in scenarios:
        print(f"\n{'=' * 20} RUNNING SCENARIO: {scenario['name']} {'=' * 20}")

```

```python
        PORTFOLIO_TYPE = scenario['portfolio']
        UNDERLYING_MODEL = scenario['model']
        BASE_HAZARD_RATE = 0.05
        n_paths_for_run = 100000

        common_params = {
            'S0': 100.0, 'K': scenario['K'], 'T_maturity': 1.0, 'r': 0.05, 'q': 0.0,
            'n_paths': n_paths_for_run, 'mpor_days': 10, 'h': 0.001, 'R': 0.4,
            'rng': np.random.default_rng(0)
        }
        gbm_params = {'sigma': 0.2}
        merton_params = {'sigma': 0.2, 'lam': 0.05, 'gamma': 0.25, 'delta_jump': scenario.get('delta_jump', 0.25)}
        heston_params = {'z0': 0.04, 'sigma': 1.0, 'kappa': 3.0, 'eta': 0.3, 'rho': scenario.get('rho', -0.7)}

        portfolio_map = {'Call': bs_call_price, 'Put': bs_put_price, 'Forward': forward_price}
        model_map = {'GBM': (generate_gbm_paths, gbm_params), 'Merton': (generate_merton_paths, merton_params),
                     'Heston': (generate_heston_paths, heston_params)}

        pricing_function = portfolio_map.get(PORTFOLIO_TYPE)
        underlying_generator, model_specific_params = model_map.get(UNDERLYING_MODEL)
        params = {**common_params, **model_specific_params, 'hazard': BASE_HAZARD_RATE}

        if UNDERLYING_MODEL == 'Heston':
            params['pricing_sigma'] = np.sqrt(params['z0'])
        else:
            params['pricing_sigma'] = params['sigma']

        delta_v_generator = partial(generate_derivative_increments, underlying_generator, pricing_function)
```

```
    im_levels = {'No IM': 0.0, 'IM @ 95%': norm.ppf(0.95), 'IM @ 99%': norm.ppf(0.99)}

    # Run Comparison Analysis for Graph
    rhos_for_graph = np.linspace(0.1, 0.9, 9)
    original_cva_results_graph = {lbl: [] for lbl in im_levels}
    for rho in rhos_for_graph:
        cva_out = calculate_cva(rho, im_levels, params, delta_v_generator)
        for lbl in im_levels: original_cva_results_graph[lbl].append(cva_out[lbl])

    cva_star_dict = calculate_cva(0.0, im_levels, params, delta_v_generator)
    CVA_STAR = cva_star_dict['No IM']


    def cva_solver(lambda_trial, rho_val, target_cva):
        current_params = params.copy();
        current_params['hazard'] = lambda_trial
        cva_out = calculate_cva(rho_val, {'No IM': 0.0}, current_params, delta_v_generator)
        return cva_out['No IM'] - target_cva


    equivalent_lambdas = []
    new_cva_results_graph = {lbl: [] for lbl in ['IM @ 95%', 'IM @ 99%']}
    for rho in rhos_for_graph:
        try:
            lambda_star = brentq(cva_solver, 1e-6, 1.0, args=(rho, CVA_STAR))
            equivalent_lambdas.append(lambda_star)
            params_new_lambda = params.copy();
            params_new_lambda['hazard'] = lambda_star
            cva_out_new = calculate_cva(rho, im_levels, params_new_lambda, delta_v_generator)
```

```
                for lbl in ['IM @ 95%', 'IM @ 99%']: new_cva_results_graph[lbl].append(cva_out_new[lbl])
            except (ValueError, RuntimeError):
                equivalent_lambdas.append(np.nan);
                for lbl in ['IM @ 95%', 'IM @ 99%']: new_cva_results_graph[lbl].append(np.nan)

        # Compile and Export Comparison Table
        results_df = pd.DataFrame({'rho_WWR': rhos_for_graph, 'Equivalent_Lambda_Star': equivalent_lambdas,
                                   'Original_CVA_95': original_cva_results_graph.get('IM @ 95%', []),
                                   'New_CVA_95': new_cva_results_graph.get('IM @ 95%', []),
                                   'Original_CVA_99': original_cva_results_graph.get('IM @ 99%', []),
                                   'New_CVA_99': new_cva_results_graph.get('IM @ 99%', [])})
        results_df['Proportion_95'] = results_df['New_CVA_95'] / results_df['Original_CVA_95']
        results_df['Proportion_99'] = results_df['New_CVA_99'] / results_df['Original_CVA_99']
        print(f"\n--- Final Comparison Table for {scenario['name']} ---");
        pd.set_option('display.float_format', '{:.5f}'.format);
        print(results_df.dropna())
        table_path = os.path.join(table_output_dir, f"{scenario['name']}_comparison_table.csv");
        results_df.to_csv(table_path)
        print(f"Comparison table saved to {table_path}")

        # Plot Graph
        fig, ax = plt.subplots(figsize=(10, 6));
        ax.plot(results_df['rho_WWR'], results_df['Proportion_95'], 'o-', label='IM @ 95% (New CVA / Original CVA)');
```

```
        ax.plot(results_df['rho_WWR'], results_df['Proportion_99'], 's-', label='IM @ 99% (New CVA / Original CVA)');
        ax.set_xlabel(r'Wrong-way correlation $\rho_{WWR}$');
        ax.set_ylabel('Proportion of Original CVA');
        ax.set_title(f"CVA Comparison - {scenario['name']}");
        ax.legend();
        ax.grid(True, linestyle=':');
        ax.set_ylim(bottom=0);
        plt.tight_layout()
        plot_path = os.path.join(plot_output_dir, f"{scenario['name']}.png");
        plt.savefig(plot_path);
        plt.close(fig)
        print(f"Plot saved to {plot_path}")

        # Calculate and Export CVA Amplification Table (Original Lambda)
        rhos_for_table = [0.0, 0.1, 0.25, 0.5, 0.75, 1.0]
        cva_table_results = {lbl: {} for lbl in im_levels}
        for rho in rhos_for_table:
            cva_out = calculate_cva(rho, im_levels, params, delta_v_generator)
            for lbl in im_levels: cva_table_results[lbl][rho] = cva_out[lbl]

        table_a_data = {}
        for label, cva_values in cva_table_results.items():
            baseline_cva = cva_values[0.0];
            if baseline_cva > 1e-9:
                ratios = [cva_values[rho] / baseline_cva for rho in rhos_for_table[1:]]
            else:
                ratios = [np.inf] * (len(rhos_for_table) - 1)
            table_a_data[label] = [baseline_cva] + ratios

        df_a = pd.DataFrame.from_dict(table_a_data, orient='index',
                                      columns=['CVA(rho=0)', 'rho=0.1', 'rho=0.25', 'rho=0.5
```

```
', 'rho=0.75', 'rho=1.0'])
        print(f"\n--- CVA Amplification Table (Original Lambda) for {scenario['name']} ---");
        df_a_display = df_a.copy();
        df_a_display['CVA(rho=0)'] = df_a_display['CVA(rho=0)'].map('{:,.4f}'.format)
        for col in df_a_display.columns[1:]: df_a_display[col] = df_a_display[col].map('{:,.1f}'.format)
        print(df_a_display)
        table_a_path = os.path.join(table_output_dir, f"{scenario['name']}_amplification_table.csv");
        df_a.to_csv(table_a_path)
        print(f"Amplification table saved to {table_a_path}")

        # Calculate and Export CVA Amplification Table (Equivalent Lambda*)
        lambda_star_table = {}
        for rho in rhos_for_table:
            if rho == 0.0:
                lambda_star_table[rho] = BASE_HAZARD_RATE
            else:
                try:
                    lambda_star_table[rho] = brentq(cva_solver, 1e-6, 1.0, args=(rho, CVA_STAR))
                except (ValueError, RuntimeError):
                    lambda_star_table[rho] = np.nan

        cva_derisked_table_results = {lbl: {} for lbl in im_levels}
        for rho in rhos_for_table:
            params_new_lambda = params.copy();
            params_new_lambda['hazard'] = lambda_star_table[rho]
            cva_out = calculate_cva(rho, im_levels, params_new_lambda, delta_v_generator)
            for lbl in im_levels: cva_derisked_table_results[lbl][rho] = cva_out[lbl]

        table_a_derisked_data = {}
        for label, cva_values in cva_derisked_table_results.items():
```

```
            baseline_cva = cva_derisked_table_results[label][0.0]
            if baseline_cva > 1e-9:
                ratios = [cva_values[rho] / baseline_cva for rho in rhos_for_table[1:]]
            else:
                ratios = [np.inf] * (len(rhos_for_table) - 1)
            table_a_derisked_data[label] = [baseline_cva] + ratios

        df_a_derisked = pd.DataFrame.from_dict(table_a_derisked_data, orient='index',
                                               columns=['CVA(rho=0)', 'rho=0.1', 'rho=0.25', 'rho=0.5', 'rho=0.75',
                                                        'rho=1.0'])
        print(f"\n--- De-Risked CVA Amplification Table (Equivalent Lambda*) for {scenario['name']} ---");
        df_a_derisked_display = df_a_derisked.copy();
        df_a_derisked_display['CVA(rho=0)'] = df_a_derisked_display['CVA(rho=0)'].map('{:,.4f}'.format)
        for col in df_a_derisked_display.columns[1:]: df_a_derisked_display[col] = df_a_derisked_display[col].map(
            '{:,.1f}'.format)
        print(df_a_derisked_display)
        table_a_derisked_path = os.path.join(table_output_dir, f"{scenario['name']}_amplification_table_derisked.csv");
        df_a_derisked.to_csv(table_a_derisked_path)
        print(f"De-Risked Amplification table saved to {table_a_derisked_path}")
```